

Graph Entropy

Information Theory Project

Niousha Rouhani, Fatemeh Mazloum Roshan

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Overview

Graph entropy, denoted as $H(G, P)$, unifies information theory and structural combinatorics by defining an information-theoretic functional on a graph G given a probability distribution P over its vertex set. While Shannon's classical entropy measures uncertainty assuming complete discernibility among alphabet symbols, real-world information sources often feature ambiguous or non-distinguishable states—such as handwritten characters or functional graphs. By modeling symbol discernibility via adjacency in a graph, graph entropy naturally generalizes Shannon's source coding theorem, determining the fundamental rate limits for data compression where indistinguishable sequence pairs share codewords within a bounded error probability. Furthermore, owing to its intrinsic sub-additivity property, graph entropy serves as a powerful analytical tool for deriving lower bounds in computational complexity, solving sorting problems, and characterizing structural properties of combinatorial objects such as perfect graphs through exact additivity.

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1 Definitions

Definition 1.1. For a subset $S \subseteq \mathbb{R}^n$, the *convex hull* of S , denoted by $\text{conv}(S)$, is the smallest convex set containing S . Equivalently:

$$\text{conv}(S) = \left\{ \sum_{i=1}^k \lambda_i x_i \mid k \in \mathbb{N}, x_i \in S, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

Definition 1.2. The vertex packing polytope $VP(G)$ of a graph G is the convex hull of the characteristic vectors of maximal independent sets of G .

Definition 1.3. A set $\mathcal{A} \subseteq \mathbb{R}_+^n$ is called a *convex corner* if it is closed, convex, has a non-empty interior, and satisfies the property that if $0 \leq a'_i \leq a_i$ for $i = 1, \dots, n$ then $\mathbf{a} \in \mathcal{A}$ implies $\mathbf{a}' \in \mathcal{A}$.

Definition 1.4. Let G be a graph on vertex set $V(G) = \{1, \dots, n\}$ and let $P = (p_1, \dots, p_n)$ be a probability distribution on $V(G)$ (i.e., $p_1 + \dots + p_n = 1$ and $p_i \geq 0$ for all i). The entropy of G with respect to P is then defined as:

$$H(G, P) = \min_{\mathbf{a} \in VP(G)} \sum_{i=1}^n p_i \log \frac{1}{a_i}.$$

Definition 1.5. Let G be a graph and let P be a probability distribution on $V(G)$. The *graph entropy* of G with respect to P is defined by:

$$H(G, P) = \min I(X; Y),$$

where the minimum is taken over all pairs of random variables (X, Y) satisfying the following conditions:

1. X takes values in $V(G)$;
2. Y takes values in the family of stable maximal sets of G ;
3. $\Pr(X \in Y) = 1$;
4. The marginal distribution of X is P .

Definition 1.6. Given a graph $G = (V, E)$, the t -th co-normal power G^t of G is defined by

$$V(G^t) = [V(G)]^t = \underbrace{V(G) \times \dots \times V(G)}_{t \text{ times}},$$

and the edge set $E(G^t)$ is given by

$$E(G^t) = \left\{ \{\mathbf{x}, \mathbf{y}\} \mid \exists i \in \{1, \dots, t\} \text{ such that } \{x_i, y_i\} \in E(G) \right\},$$

where $\mathbf{x} = (x_1, \dots, x_t)$ and $\mathbf{y} = (y_1, \dots, y_t)$ are vertices in $V(G^t)$.

Definition 1.7. Let $G = (V, E)$ be a graph and let P be a probability distribution on the vertex set V . The *graph entropy* of G with respect to P , is defined as

$$H(G, P) = \lim_{t \rightarrow \infty} \min_{\substack{U \subseteq V^t \\ P^t(U) > 1-\varepsilon}} \frac{1}{t} \log_2 \chi(G^t(U)),$$

for a fix $0 < \varepsilon < 1$, where

$$P^t(U) = \sum_{\mathbf{x} \in U} P^t(\mathbf{x}),$$

for any sequence $\mathbf{x} = (x_1, \dots, x_t) \in V^t$, the product distribution $P^t(\mathbf{x})$ is given by

$$P^t(\mathbf{x}) = \prod_{i=1}^t P(x_i),$$

and $\chi(H)$ denotes the chromatic number of a graph H .

Definition 1.8. Let $G = (V, E)$ be a graph and let P be a probability distribution on the vertex set V . Then, the cartesian power G_π^t of G , is defined as the following graph:

$$V(G_\pi^t) = V(G^t),$$

and

$$E(G_\pi^t) = \left\{ \{\mathbf{x}, \mathbf{y}\} \mid \forall i \in \{1, \dots, t\} \text{ either } \{x_i, y_i\} \in E(G) \text{ or } x_i = y_i \right\},$$

where $\mathbf{x} = (x_1, \dots, x_t)$ and $\mathbf{y} = (y_1, \dots, y_t)$ are vertices in $V(G_\pi^t)$. Also, for any sequence $\mathbf{x} = (x_1, \dots, x_t) \in V(G_\pi^t)$,

$$P_\pi^t(\mathbf{x}) = \prod_{i=1}^t P(x_i) = P^t(\mathbf{x}).$$

Note that, $\overline{G^t} = (\overline{G})_\pi^t$, and $\overline{G_\pi^t} = (\overline{G})^t$.

Definition 1.9. A *kernel* is a maximal independent set of vertices of G , i.e. $A \subseteq V(G)$ is a kernel if

- (i) no two vertices of A are connected in G ,
- (ii) each $x \in V(G) - A$ is connected to at least one $y \in A$.

Definition 1.10. Given a finite set X and a probability distribution P on it, a sequence $x \in X^n$ is called *P-typical* if for each $y \in X$, the number of occurrences of the element y in $\mathbf{x}$, $N(y \mid \mathbf{x})$, satisfies the inequality

$$|N(y \mid \mathbf{x}) - nP(y)| \leq K\sqrt{nP(y)},$$

for a constant K .

Definition 1.11. Let $G = (V, E)$ be a graph, and let $S(G)$ denote the family of all maximal independent sets of G . The *fractional chromatic number* of G , denoted by $\chi_f(G)$, is defined as the following:

$$\chi_f(G) = \min \sum_{S \in S(G)} w_S,$$

subject to

$$\sum_{\substack{S \in S(G) \\ v \in S}} w_S \geq 1 \quad (\forall v \in V), \quad w_S \geq 0 \quad (\forall S \in S(G)).$$

2 Equivalence of Graph Entropy Definitions

Lemma 2.1. *Let $S(\cdot)$ denote the family of kernels. Then, $S(G^n) = S(G)^n$.*

Proof. Let $\mathbf{Y} = (Y_1, \dots, Y_n) \in S(G)^n$ choose arbitrary. Independence of $\mathbf{Y}$ follows directly from Definition 1.6 and Y_i being independent sets in G . To prove maximality, let $\mathbf{x} \notin \mathbf{Y}$. Then, there exists $k \in \{1, \dots, n\}$ such that $x_k \notin Y_k$. Since Y_k is a kernel of G , x_k must be adjacent to a vertex $y_k^* \in Y_k$ in G . Now, consider $\mathbf{y} = (y_1, \dots, y_n) \in \mathbf{Y}$, where $y_k = y_k^*$, and y_i be any arbitrary vertex in Y_i for $i \neq k$. This results in $\mathbf{x}$ being adjacent to $\mathbf{y} \in \mathbf{Y}$, proving $\mathbf{Y}$ is a maximal independent set of G^n .

To prove the reverse, let $\mathbf{Y} \in S(G^n)$, and $Y_1, \dots, Y_n$ be all vertices in its i th coordinate. The independence of Y_i is obvious. Now, suppose, for contradiction, that Y_i is not a kernel of G for some $i \in 1, \dots, n$. Then, there exists a vertex $y_i \in V(G) \setminus Y_i$ such that y_i is not adjacent to any vertex in Y_i . Consider a vertex $\mathbf{y} = (y_1, \dots, y_n) \in V(G^n)$ whose i -th coordinate is y_i , and whose other coordinates y_j are arbitrary vertices in Y_j for $j \neq i$. By the definition 1.6, $\mathbf{y}$ is not adjacent to any vertex in $\mathbf{Y}$. This contradicts the fact that $\mathbf{Y}$ is a kernel of G^n . Therefore, Y_i must be a kernel of G . Since i was arbitrary, we conclude that $Y_i \in S(G)$ for every $i \in 1, \dots, n$. Also, since $\mathbf{Y}$ is a kernel, it should be $Y_1 \times \dots \times Y_n$. Hence, $S(G^n) = S(G)^n$. $\square$

Lemma 2.2. *The chromatic number $\chi(G)$ of an arbitrary graph equals the minimum number of kernels covering G .*

Proof. Let us denote the minimum number of covering kernels by C . Then, obviously, $C \leq \chi(G)$, since the components of an arbitrary coloring of G can be extended to kernels. On the other hand, let us consider a covering kernel system $[A_1, A_2, \dots, A_C]$ with a minimum number of kernels. Because of the minimality of this covering, each of its kernels contains at least one vertex of G not contained in any other kernel of the system. Thus we obtain disjoint sets of vertices $A_1, A_2 \setminus A_1, A_3 \setminus (A_1 \cup A_2), \dots, A_C \setminus \bigcup_{i=1}^{C-1} A_i$ providing a coloring of G , which consists of exactly C components. Thus $\chi(G) = C$. $\square$

Lemma 2.3. *Let $T^n(P)$ denote the set of P -typical n -sequences, defined in Definition 1.10. Then:*

(i) $\forall \lambda, \exists K$ such that for this K ,

$$P(\overline{T^n(\mathbf{P})}) < \lambda,$$

where

$$P(\overline{T^n(\mathbf{P})}) = \sum_{x \in \overline{T^n(\mathbf{P})}} P^n(x),$$

and $\overline{S}$ denotes the complement of the set S . (Every $K > \sqrt{|X|/\lambda}$ is suitable).

(ii) For every $x \in T^n(P)$,

$$2^{-(nH(P)+C\sqrt{n})} \leq P^n(x) \leq 2^{-(nH(P)-C\sqrt{n})}.$$

(iii) For every $n \in \mathbb{N}$,

$$2^{nH(P)-C\sqrt{n}} \leq |T^n(P)| \leq 2^{nH(P)+C\sqrt{n}}.$$

The constant C appearing in both (ii) and (iii) does not depend on n and P ; it depends only on $|X|$ and λ .

Proof.

Proof of (i):

For each $y \in X$, the count $N(y | \mathbf{x})$ is a binomial random variable with mean $\mu = n\mathbf{P}(y)$ and variance $\sigma^2 = n\mathbf{P}(y)(1 - \mathbf{P}(y))$. Since $1 - \mathbf{P}(y) \leq 1$, we bound the variance as $\sigma^2 \leq n\mathbf{P}(y)$.

Chebyshev's inequality states that for any random variable Z with mean μ and variance σ^2 , and for any $k > 0$:

$$\mathbf{P}(|Z - \mu| > k) \leq \frac{\sigma^2}{k^2}$$

If we put $Z = N(y | \mathbf{x})$, average value $\mu = n\mathbf{P}(y)$, and distance limit $k = K\sqrt{n\mathbf{P}(y)}$, the probability that the difference is larger than $K\sqrt{n\mathbf{P}(y)}$ for a single symbol $y \in X$ is at most:

$$\mathbf{P}(|N(y | \mathbf{x}) - n\mathbf{P}(y)| > K\sqrt{n\mathbf{P}(y)}) \leq \frac{\sigma^2}{(K\sqrt{n\mathbf{P}(y)})^2} = \frac{\sigma^2}{K^2 n\mathbf{P}(y)} \leq \frac{n\mathbf{P}(y)}{K^2 n\mathbf{P}(y)} = \frac{1}{K^2}$$

A sequence $\mathbf{x}$ is non-typical (i.e., $\mathbf{x} \in \overline{T^n(\mathbf{P})}$) if this condition fails for at least one symbol $y \in X$. Summing these probabilities over all $|X|$ symbols in the alphabet yields:

$$\mathbf{P}(\overline{T^n(\mathbf{P})}) \leq \sum_{y \in X} \mathbf{P}(|N(y | \mathbf{x}) - n\mathbf{P}(y)| > K\sqrt{n\mathbf{P}(y)}) \leq \sum_{y \in X} \frac{1}{K^2} = \frac{|X|}{K^2}$$

To ensure that this total probability remains strictly less than λ , we set $\frac{|X|}{K^2} < \lambda$, which gives $K^2 > \frac{|X|}{\lambda}$ and consequently $K > \sqrt{\frac{|X|}{\lambda}}$.

Proof of (ii):

According to Definition 1.10, for any typical sequence $\mathbf{x} \in T^n(\mathbf{P})$, the occurrence count $N(y | \mathbf{x})$ for each symbol $y \in X$ satisfies:

$$n\mathbf{P}(y) - K\sqrt{n\mathbf{P}(y)} \leq N(y | \mathbf{x}) \leq n\mathbf{P}(y) + K\sqrt{n\mathbf{P}(y)}$$

Since $\sqrt{\mathbf{P}(y)} \leq 1$ for all $y \in X$, these bounds simplify to:

$$n\mathbf{P}(y) - K\sqrt{n} \leq N(y | \mathbf{x}) \leq n\mathbf{P}(y) + K\sqrt{n}$$

Due to the independence of the components in sequence $\mathbf{x}$, the overall probability $\mathbf{P}(\mathbf{x})$ equals the product of the individual probabilities, which can be rewritten in terms of symbol frequencies as:

$$\mathbf{P}(\mathbf{x}) = \prod_{i=1}^n \mathbf{P}(x_i) = \prod_{y \in X} \mathbf{P}(y)^{N(y|\mathbf{x})}$$

Substituting the lower and upper bounds of $N(y | \mathbf{x})$ into the product yields:

$$\prod_{y \in X} \mathbf{P}(y)^{n\mathbf{P}(y) + K\sqrt{n}} \leq \mathbf{P}(\mathbf{x}) \leq \prod_{y \in X} \mathbf{P}(y)^{n\mathbf{P}(y) - K\sqrt{n}}$$

Taking the logarithm transforms the product into a summation:

$$\sum_{y \in X} (n\mathbf{P}(y) + K\sqrt{n}) \log_2 \mathbf{P}(y) \leq \log_2 \mathbf{P}(\mathbf{x}) \leq \sum_{y \in X} (n\mathbf{P}(y) - K\sqrt{n}) \log_2 \mathbf{P}(y)$$

Expanding the logarithmic summation gives the following inequality:

$$\sum_{y \in X} n\mathbf{P}(y) \log_2 \mathbf{P}(y) + K\sqrt{n} \sum_{y \in X} \log_2 \mathbf{P}(y) \leq \log_2 \mathbf{P}(\mathbf{x}) \leq \sum_{y \in X} n\mathbf{P}(y) \log_2 \mathbf{P}(y) - K\sqrt{n} \sum_{y \in X} \log_2 \mathbf{P}(y)$$

By the definition of Shannon entropy, $H(\mathbf{P}) = -\sum_{y \in X} \mathbf{P}(y) \log_2 \mathbf{P}(y)$, the first term simplifies to $-nH(\mathbf{P})$. Furthermore, since the alphabet X is finite, the above bound holds for any constant C (independent of n) satisfying:

$$C \geq K \sum_{y \in X} |\log_2 \mathbf{P}(y)|$$

Substituting these terms into the inequality yields the logarithmic bounds:

$$-nH(\mathbf{P}) - C\sqrt{n} \leq \log_2 \mathbf{P}(\mathbf{x}) \leq -nH(\mathbf{P}) + C\sqrt{n}$$

Exponentiating all terms with base 2 completes the proof:

$$2^{-(nH(\mathbf{P}) + C\sqrt{n})} \leq \mathbf{P}(\mathbf{x}) \leq 2^{-(nH(\mathbf{P}) - C\sqrt{n})}$$

Proof of (iii):

To find the upper bound for $|T^n(\mathbf{P})|$, notice that the total probability of all typical sequences in $T^n(\mathbf{P})$ cannot be more than 1:

$$\sum_{\mathbf{x} \in T^n(\mathbf{P})} \mathbf{P}(\mathbf{x}) = \mathbf{P}(T^n(\mathbf{P})) \leq 1$$

Using the lower bound for sequence probability $\mathbf{P}(\mathbf{x})$ from part (ii), we get:

$$\sum_{\mathbf{x} \in T^n(\mathbf{P})} 2^{-(nH(\mathbf{P}) + (K \sum_{y \in X} |\log_2 \mathbf{P}(y)|) \sqrt{n})} \leq \sum_{\mathbf{x} \in T^n(\mathbf{P})} \mathbf{P}(\mathbf{x}) \leq 1$$

From part (ii), we know that any constant $C \geq K \sum_{y \in X} |\log_2 \mathbf{P}(y)|$ works for this inequality. Since the exponent does not depend on $\mathbf{x}$, we can factor it out of the sum and multiply both sides by the exponential term to get:

$$|T^n(P)| \leq 2^{nH(\mathbf{P}) + (K \sum_{y \in X} |\log_2 \mathbf{P}(y)|)\sqrt{n}}$$

For the lower bound, part (i) tells us that the probability of non-typical sequences is strictly less than λ . So the total probability of typical sequences must be at least $1 - \lambda$:

$$1 - \lambda \leq \mathbf{P}(T^n(\mathbf{P})) = \sum_{\mathbf{x} \in T^n(\mathbf{P})} \mathbf{P}(\mathbf{x})$$

Plugging in the upper bound for $\mathbf{P}(\mathbf{x})$ from part (ii) and multiplying both sides by the exponential term gives:

$$|T^n(P)| \geq (1 - \lambda) \cdot 2^{nH(\mathbf{P}) - (K \sum_{y \in X} |\log_2 \mathbf{P}(y)|)\sqrt{n}}$$

Rewriting $(1 - \lambda)$ as $2^{\log_2(1-\lambda)}$, we have:

$$|T^n(P)| \geq 2^{nH(\mathbf{P}) - (K \sum_{y \in X} |\log_2 \mathbf{P}(y)|)\sqrt{n} + \log_2(1-\lambda)}$$

To match both bounds using the same constant C , we set:

$$-C\sqrt{n} \leq - \left(K \sum_{y \in X} |\log_2 \mathbf{P}(y)| \right) \sqrt{n} + \log_2(1 - \lambda)$$

Multiplying by -1 and using $-\log_2(1 - \lambda) = |\log_2(1 - \lambda)|$:

$$C\sqrt{n} \geq \left(K \sum_{y \in X} |\log_2 \mathbf{P}(y)| \right) \sqrt{n} + |\log_2(1 - \lambda)|$$

Dividing both sides by $\sqrt{n}$ gives:

$$C \geq K \sum_{y \in X} |\log_2 \mathbf{P}(y)| + \frac{|\log_2(1 - \lambda)|}{\sqrt{n}}$$

Since $\frac{1}{\sqrt{n}} \leq 1$ for all $n \geq 1$, we can choose any constant C such that:

$$C \geq K \sum_{y \in X} |\log_2 \mathbf{P}(y)| + |\log_2(1 - \lambda)|$$

This choice of C satisfies part (ii) and handles the $(1 - \lambda)$ term for all $n \geq 1$, which completes the proof:

$$2^{nH(\mathbf{P}) - C\sqrt{n}} \leq |T^n(P)| \leq 2^{nH(\mathbf{P}) + C\sqrt{n}}$$

□

The following notation is used throughout Lemmas 2.4, 2.5, and 2.6. Let X be a finite alphabet partitioned into disjoint equivalence classes, denoted by $E(X)$. For a given probability distribution P on X , the probability of a class $e \in E(X)$ is defined as:

$$P(e) = \sum_{y \in e} P(y).$$

This naturally induces the class distribution P_E . Moreover, for any sequence $\mathbf{x} = (x_1, \dots, x_n) \in X^n$, its corresponding class sequence is denoted by:

$$\mathbf{e}(\mathbf{x}) = (e(x_1), \dots, e(x_n)),$$

where $e(x_i) \in E(X)$ represents the specific class containing x_i .

Lemma 2.4. *If $\mathbf{x} \in T^n(\mathbf{P})$, then its equivalence class satisfies $\mathbf{e}(\mathbf{x}) \in T^n(\mathbf{P}_E)$.*

Proof. The occurrence count of any equivalence class $e \in E(X)$ in the sequence $\mathbf{e}(\mathbf{x})$, denoted by $N(e | \mathbf{e})$, is given by the sum of the occurrences of its constituent symbols:

$$N(e | \mathbf{e}) = \sum_{y \in e} N(y | \mathbf{x}).$$

Since $\mathbf{x}$ is typical, the count of any element y is very close to its expected value:

$$|N(y | \mathbf{x}) - nP(y)| \leq K_0 \sqrt{nP(y)}$$

To find the bound for an entire class e , we simply add up the differences for all elements inside that class. Using the basic triangle inequality, we get:

$$\begin{aligned} |N(e | \mathbf{e}) - nP(e)| &= \left| \sum_{y \in e} (N(y | \mathbf{x}) - nP(y)) \right| \\ &\leq \sum_{y \in e} |N(y | \mathbf{x}) - nP(y)| \\ &\leq K_0 \sqrt{n} \sum_{y \in e} \sqrt{P(y)} \end{aligned}$$

To bound the sum, we use the Cauchy-Schwarz inequality:

$$\sum_{y \in e} 1 \cdot \sqrt{P(y)} \leq \sqrt{\sum_{y \in e} 1} \cdot \sqrt{\sum_{y \in e} P(y)} = \sqrt{|e|} \sqrt{P(e)}$$

Putting this back into our earlier result, we get:

$$|N(e | \mathbf{e}) - nP(e)| \leq (K_0 \sqrt{|e|}) \sqrt{nP(e)}$$

Now, we can define a new constant K_1 that covers the largest possible class:

$$K_1 = K_0 \max_e \sqrt{|e|}$$

Since K_1 only depends on the classes (and not on n), our bound perfectly matches the definition of a typical sequence:

$$|N(e | \mathbf{e}) - nP(e)| \leq K_1 \sqrt{nP(e)} \implies \mathbf{e}(\mathbf{x}) \in T^n(\mathbf{P}_E) \quad \blacksquare$$

□

Lemma 2.5. Let $\mathbf{x} \in X^n$ be a sequence with class sequence $\mathbf{e}$, and let $\mathbf{x}_e$ denote its restricted subsequence to elements of class $e \in E(X)$. The sequence $\mathbf{x}$ is typical ($\mathbf{x} \in T^n(\mathbf{P})$) if and only if for every class $e \in E(X)$, its restricted subsequence $\mathbf{x}_e$ is typical with respect to the conditional distribution $\mathbf{P}_e = \mathbf{P}(\cdot | e)$:

$$\mathbf{x} \in T^n(\mathbf{P}) \iff \forall e \in E(X), \quad \mathbf{x}_e \in T^{N(e | \mathbf{e})}(\mathbf{P}_e).$$

Proof. Since $\mathbf{e} \in T^n(\mathbf{P}_E)$, the number of occurrences $N(e | \mathbf{e})$ for each class $e \in E(X)$ satisfies the typicality bound in Definition 1.10:

$$|N(e | \mathbf{e}) - nP(e)| \leq K_1 \sqrt{nP(e)}.$$

for some constant K_1 . For any symbol $y \in e$, its count in the full sequence equals its count in the restricted subsequence, giving $N(y | \mathbf{x}) = N(y | \mathbf{x}_e)$. Using the relation $P(y) = P(e)P(y | e)$, we bound the difference between $nP(y)$ and $N(e | \mathbf{e})P(y | e)$ as follows:

$$|nP(y) - N(e | \mathbf{e})P(y | e)| = |N(e | \mathbf{e}) - nP(e)|P(y | e) \leq K_1 \sqrt{nP(e)}P(y | e) = K_1 \sqrt{nP(y)P(y | e)}. \quad (1)$$

To prove the forward direction, assume $\mathbf{x} \in T^n(\mathbf{P})$. By Definition 1.10, we have $|N(y | \mathbf{x}) - nP(y)| \leq K_0 \sqrt{nP(y)}$ for some constant K_0 . Combining this with (1) via the triangle inequality yields:

$$\begin{aligned} |N(y | \mathbf{x}_e) - N(e | \mathbf{e})P(y | e)| &= |N(y | \mathbf{x}) - nP(y) + nP(y) - N(e | \mathbf{e})P(y | e)| \\ &\leq |N(y | \mathbf{x}) - nP(y)| + |nP(y) - N(e | \mathbf{e})P(y | e)| \\ &\leq K_0 \sqrt{nP(y)} + K_1 \sqrt{nP(y)P(y | e)} \\ &\leq (K_0 + K_1) \sqrt{nP(y)}, \end{aligned}$$

To complete the proof for the forward direction, we must express our upper bound entirely in terms of the restricted sequence's length, $N(e | \mathbf{e})$. From the typicality of $\mathbf{e}$, we have $nP(e) - N(e | \mathbf{e}) \leq K_1 \sqrt{nP(e)}$. Treating this as a quadratic inequality of the form $X^2 - K_1 X - N(e | \mathbf{e}) \leq 0$ where $X = \sqrt{nP(e)}$, we solve for its positive root. Using the sub-additivity of the square root ($\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$), we can bound X step-by-step:

$$\begin{aligned} \sqrt{nP(e)} &\leq \frac{K_1 + \sqrt{K_1^2 + 4N(e | \mathbf{e})}}{2} \\ &\leq \frac{K_1 + \sqrt{K_1^2} + \sqrt{4N(e | \mathbf{e})}}{2} \\ &= \frac{2K_1 + 2\sqrt{N(e | \mathbf{e})}}{2} \\ &= K_1 + \sqrt{N(e | \mathbf{e})} \\ &\leq (1 + K_1) \sqrt{N(e | \mathbf{e})}, \end{aligned}$$

where the final step assumes the non-trivial case $N(e | \mathbf{e}) \geq 1$. Multiplying this chain of inequalities by $\sqrt{P(y | e)}$ and recalling that $P(e)P(y | e) = P(y)$, we find:

$$\sqrt{nP(y)} \leq (1 + K_1) \sqrt{N(e | \mathbf{e})P(y | e)}.$$

Substituting this upper bound back into our previous triangle inequality result, we obtain the final condition:

$$|N(y | \mathbf{x}_e) - N(e | \mathbf{e})P(y | e)| \leq (K_0 + K_1)(1 + K_1) \sqrt{N(e | \mathbf{e})P(y | e)}.$$

Setting $K_e = (K_0 + K_1)(1 + K_1)$ provides a constant bound independent of the subsequence length $N(e | \mathbf{e})$. This confirms that the restricted subsequence meets the requirements of Definition 1.10, concluding that $\mathbf{x}_e \in T^{N(e | \mathbf{e})}(\mathbf{P}_e)$.

To prove the converse direction, suppose that $\mathbf{x}_e \in T^{N(e | \mathbf{e})}(\mathbf{P}_e)$ for all $e \in E(X)$. By Definition 1.10, there exists a constant K_e such that $|N(y | \mathbf{x}_e) - N(e | \mathbf{e})P(y | e)| \leq K_e \sqrt{N(e | \mathbf{e})P(y | e)}$. Recall that for any $y \in e$, $N(y | \mathbf{x}) = N(y | \mathbf{x}_e)$. By substituting this into our target deviation, adding and subtracting $N(e | \mathbf{e})P(y | e)$ inside the absolute value, and applying the triangle inequality along with (1), we obtain:

$$\begin{aligned} |N(y | \mathbf{x}) - nP(y)| &= |N(y | \mathbf{x}_e) - N(e | \mathbf{e})P(y | e) + N(e | \mathbf{e})P(y | e) - nP(y)| \\ &\leq |N(y | \mathbf{x}_e) - N(e | \mathbf{e})P(y | e)| + |nP(y) - N(e | \mathbf{e})P(y | e)| \\ &\leq K_e \sqrt{N(e | \mathbf{e})P(y | e)} + K_1 \sqrt{nP(y)P(y | e)}. \end{aligned}$$

To unify the bounds, note that $N(e | \mathbf{e}) \leq n$. Letting $P_{\min} = \min_{e \in E(X)} P(e) > 0$ and recalling that $P(y | e) = P(y)/P(e) \leq P(y)/P_{\min}$, we can bound the first square root term as:

$$\sqrt{N(e | \mathbf{e})P(y | e)} \leq \sqrt{n \frac{P(y)}{P_{\min}}} = \frac{1}{\sqrt{P_{\min}}} \sqrt{nP(y)}.$$

Substituting this back, and noting that $P(y | e) \leq 1$ for the second term, we obtain the final global bound:

$$|N(y | \mathbf{x}) - nP(y)| \leq \left(\frac{K_e}{\sqrt{P_{\min}}} + K_1 \right) \sqrt{nP(y)}.$$

Setting $K = K_e/\sqrt{P_{\min}} + K_1$, this matches the exact structure of Definition 1.10, thereby completing the proof that $\mathbf{x} \in T^n(\mathbf{P})$. $\square$

Lemma 2.6. *Let $\mathbf{e} \in T^n(P_E)$ be a typical sequence of equivalence classes. The number of typical sequences contained within the clique $\mathbf{e}$ satisfies:*

$$2^{nH(P|\mathbf{e})-C\sqrt{n}} \leq |T^n(P) \cap \mathbf{e}| \leq 2^{nH(P|\mathbf{e})+C\sqrt{n}},$$

where $C > 0$ is a numerical constant independent of n .

Proof. Based on Lemma 2.5, a sequence $\mathbf{x}$ is in $T^n(\mathbf{P}) \cap \mathbf{e}$ if and only if its restricted subsequence $\mathbf{x}_e$ is conditionally typical for all $e \in E(X)$.

Because selections across disjoint classes are independent, the size of the intersection $|T^n(\mathbf{P}) \cap \mathbf{e}|$ is the product of the sizes of these individual sets. Applying Lemma 2.3 to each class gives:

$$\begin{aligned} |T^n(\mathbf{P}) \cap \mathbf{e}| &= \prod_{e \in E(X)} |T^{N(e|\mathbf{e})}(\mathbf{P}_e)| \\ &\geq \prod_{e \in E(X)} 2^{N(e|\mathbf{e})H(\mathbf{P}_e) - K_e\sqrt{N(e|\mathbf{e})}} \\ &= 2^{\sum_{e \in E(X)} N(e|\mathbf{e})H(\mathbf{P}_e) - \sum_{e \in E(X)} K_e\sqrt{N(e|\mathbf{e})}}. \end{aligned}$$

To evaluate the exponent, we write $N(e|\mathbf{e})$ as $nP(e) + (N(e|\mathbf{e}) - nP(e))$. This separates the expected value from the deviation term:

$$\begin{aligned} \sum_{e \in E(X)} N(e|\mathbf{e})H(\mathbf{P}_e) &= \sum_{e \in E(X)} nP(e)H(\mathbf{P}_e) + \sum_{e \in E(X)} (N(e|\mathbf{e}) - nP(e))H(\mathbf{P}_e) \\ &= nH(P|\mathbf{e}) + \sum_{e \in E(X)} (N(e|\mathbf{e}) - nP(e))H(\mathbf{P}_e). \end{aligned}$$

From Definition 1.10, we know that $|N(e|\mathbf{e}) - nP(e)| \leq K_1\sqrt{nP(e)}$. Using the triangle inequality, we can bound the deviation term:

$$\begin{aligned} \left| \sum_{e \in E(X)} (N(e|\mathbf{e}) - nP(e))H(\mathbf{P}_e) \right| &\leq \sum_{e \in E(X)} |N(e|\mathbf{e}) - nP(e)|H(\mathbf{P}_e) \\ &\leq K_1\sqrt{n} \sum_{e \in E(X)} \sqrt{P(e)}H(\mathbf{P}_e). \end{aligned}$$

Also, since $N(e|\mathbf{e}) \leq n$, the second part of the exponent satisfies:

$$\sum_{e \in E(X)} K_e\sqrt{N(e|\mathbf{e})} \leq \sqrt{n} \sum_{e \in E(X)} K_e.$$

To satisfy Lemma 2.3, our constant must be independent of the distribution $\mathbf{P}$. Since $\sqrt{P(e)} \leq 1$ and the entropy is bounded by $H(\mathbf{P}_e) \leq \log_2 \|X\|$, we can factor these out. By letting $K_{\max} = \max_{e \in E(X)} K_e$, we define the constant C_1 simply as:

$$C_1 = K_1\|E(X)\| \log_2 \|X\| + \|E(X)\|K_{\max}.$$

Using this constant, we can combine the bounds to get the final lower limit:

$$|T^n(\mathbf{P}) \cap \mathbf{e}| \geq 2^{nH(P|\mathbf{e}) - C_1\sqrt{n}}.$$

To find the upper bound, we consider the conditional probability of any typical sequence $\mathbf{x} \in T^n(\mathbf{P}) \cap \mathbf{e}$. Taking the logarithm gives:

$$\log_2 P(\mathbf{x}|\mathbf{e}) = \sum_{y \in X} N(y|\mathbf{x}) \log_2 P(y|e(y)).$$

By writing $N(y \mid \mathbf{x})$ as $nP(y) + (N(y \mid \mathbf{x}) - nP(y))$, we separate the expected value from the error term:

$$\log_2 P(\mathbf{x} \mid \mathbf{e}) = -nH(P \mid e) + \sum_{y \in X} (N(y \mid \mathbf{x}) - nP(y)) \log_2 P(y \mid e(y)).$$

Since $\mathbf{x}$ is a $\mathbf{P}$ -typical sequence, Definition 1.10 states that $|N(y \mid \mathbf{x}) - nP(y)| \leq K\sqrt{nP(y)}$. Applying the triangle inequality to the sum, we get:

$$\begin{aligned} \left| \sum_{y \in X} (N(y \mid \mathbf{x}) - nP(y)) \log_2 P(y \mid e(y)) \right| &\leq \sum_{y \in X} K\sqrt{nP(y)} |\log_2 P(y \mid e(y))| \\ &= \sqrt{n} \left(K \sum_{y \in X} \sqrt{P(y)} |\log_2 P(y \mid e(y))| \right). \end{aligned}$$

To ensure our constant satisfies Lemma 2.3, it must not depend on the distribution $\mathbf{P}$. Notice that $P(y) \leq P(y \mid e(y))$. If we let $u = P(y \mid e(y))$, each term in the sum is bounded by $\sqrt{u} |\log_2 u|$. The function $f(u) = \sqrt{u} |\log_2 u|$ has a fixed maximum value M on the interval $(0, 1]$. Therefore, the entire sum is bounded by $\|X\|M$.

We can now define the constant C_2 simply as:

$$C_2 = K\|X\|M.$$

This constant C_2 depends only on the alphabet size $\|X\|$ and the parameter K , which perfectly matches the requirement in Lemma 2.3. Using this constant, we can bound the conditional probability:

$$\log_2 P(\mathbf{x} \mid \mathbf{e}) \geq -nH(P \mid e) - C_2\sqrt{n}.$$

This implies:

$$P(\mathbf{x} \mid \mathbf{e}) \geq 2^{-nH(P \mid e) - C_2\sqrt{n}}.$$

Since the sum of probabilities for all sequences in $T^n(\mathbf{P}) \cap \mathbf{e}$ is at most 1, we obtain:

$$|T^n(\mathbf{P}) \cap \mathbf{e}| \cdot 2^{-nH(P \mid e) - C_2\sqrt{n}} \leq 1 \implies |T^n(\mathbf{P}) \cap \mathbf{e}| \leq 2^{nH(P \mid e) + C_2\sqrt{n}}.$$

Finally, setting a single constant $C = \max(C_1, C_2)$ satisfies both bounds and completes the proof. $\square$

Lemma 2.7. *If $x \sim y \Leftrightarrow \{ \text{either } (x, y) \in E(X) \text{ or } x = y \}$ is an equivalence relation on the vertex set (i.e. G is the union of pairwise disjoint complete subgraphs), then*

$$2^{nH(\mathbf{P} \mid e) - K\sqrt{n}} \leq \chi_\pi(n, \lambda) \leq 2^{nH(\mathbf{P} \mid e) + K\sqrt{n}}$$

where

$$H(\mathbf{P} \mid e) = \sum_{x \in X} P(x) \log_2 \frac{P(e(x))}{P(x)}$$

and $e(x)$ denotes the union of the elements equivalent to x . K is a numerical constant independent of n and $\mathbf{P}$ but dependent of $\|X\|$ and λ .

Proof. We establish the lower and upper bounds separately.

To prove the lower bound, let $S \subseteq X^n$ be the optimal subset, such that $P(S) \geq 1 - \lambda$ and $\chi(G^n[S]) = \chi_\pi(n, \lambda)$. By Lemma 2.3, setting $\epsilon = \frac{1-\lambda}{2}$ ensures the existence of a constant K such that $P(T^n(\mathbf{P})) < \epsilon$. This yields the strictly positive typical intersection bound:

$$P(S \cap T^n(\mathbf{P})) \geq P(S) - P(\overline{T^n(\mathbf{P})}) \geq 1 - \lambda - \frac{1 - \lambda}{2} = \frac{1 - \lambda}{2} = \delta > 0.$$

According to Lemma 2.3, for every typical sequence $\mathbf{x} \in S \cap T^n(\mathbf{P})$, the probability is upper-bounded by $P(\mathbf{x}) \leq 2^{-nH(\mathbf{P}) + C_1\sqrt{n}}$. Therefore:

$$\delta \leq P(S \cap T^n(\mathbf{P})) \leq |S \cap T^n(\mathbf{P})| \cdot 2^{-nH(\mathbf{P}) + C_1\sqrt{n}},$$

which gives an initial lower bound on the cardinality of the intersection:

$$|S \cap T^n(\mathbf{P})| \geq \delta \cdot 2^{nH(\mathbf{P}) - C_1\sqrt{n}} = 2^{nH(\mathbf{P}) - C_1\sqrt{n} + \log_2 \delta}.$$

Since $0 < \delta < 1$, we have $\log_2 \delta < 0$. For any $n \geq 1$, the inequality $\log_2 \delta \geq \sqrt{n} \log_2 \delta$ holds. Thus, we can bound the cardinality independently of n in the constant term:

$$|S \cap T^n(\mathbf{P})| \geq 2^{nH(\mathbf{P}) - (C_1 - \log_2 \delta)\sqrt{n}} = 2^{nH(\mathbf{P}) - C'_1\sqrt{n}},$$

where $C'_1 = C_1 - \log_2 \delta$. Because C_1 depends only on $\|X\|$ and λ (as per Lemma 2.3), and δ depends solely on λ , the new constant C'_1 is strictly independent of n and $\mathbf{P}$.

Next, consider any arbitrary independent set I in G^n . Because the base graph G is a disjoint union of cliques, G^n is equivalently partitioned into cliques $C(\mathbf{e})$ indexed by the equivalence class sequences. Since an independent set can contain at most one vertex from any clique, $|I \cap C(\mathbf{e})| \leq 1$. Consequently, According to Lemma 2.4 the number of vertices an independent set can have within the typical set is bounded by the total number of typical equivalence class sequences. Applying Lemma 2.3 to the equivalence class distribution $\mathbf{P}_E$, we obtain:

$$|I \cap S \cap T^n(\mathbf{P})| \leq |T^n(\mathbf{P}_E)| \leq 2^{nH(\mathbf{P}_E) + C_2\sqrt{n}}.$$

Since $S \cap T^n(\mathbf{P}) \subseteq S$, the total chromatic number is bounded by the ratio of the subset size to the maximum independent set size:

$$\begin{aligned} \chi_\pi(n, \lambda) &\geq \frac{|S \cap T^n(\mathbf{P})|}{\max_I |I \cap S \cap T^n(\mathbf{P})|} \\ &\geq \frac{2^{nH(\mathbf{P}) - C'_1\sqrt{n}}}{2^{nH(\mathbf{P}_E) + C_2\sqrt{n}}} \\ &= 2^{n(H(\mathbf{P}) - H(\mathbf{P}_E)) - (C'_1 + C_2)\sqrt{n}}. \end{aligned}$$

By substituting the entropy identity $H(\mathbf{P}) - H(\mathbf{P}_E) = H(\mathbf{P} | e)$, we establish the lower bound:

$$\chi_\pi(n, \lambda) \geq 2^{nH(\mathbf{P}|e) - K_1\sqrt{n}},$$

where $K_1 = C'_1 + C_2$ is independent of n and $\mathbf{P}$.

To establish the upper bound, Lemma 2.3 guarantees that there exists a constant $K > \sqrt{\|X\|/\lambda}$ such that $P(T^n(\mathbf{P})) < \lambda$. Thus, the typical set itself satisfies the required probability threshold, $P(T^n(\mathbf{P})) \geq 1 - \lambda$, providing the upper bound:

$$\chi_\pi(n, \lambda) \leq \chi(G^n[T^n(\mathbf{P})]).$$

Based on Definition 1.10, if a sequence is typical, its corresponding equivalence class sequence is also typical:

$$\mathbf{x} \in T^n(\mathbf{P}) \implies \mathbf{e} = e(\mathbf{x}) \in T^n(\mathbf{P}_E).$$

This ensures that all vertices in $T^n(\mathbf{P})$ belong exclusively to typical equivalence classes. Therefore, the induced subgraph $G^n[T^n(\mathbf{P})]$ is a disjoint union of cliques $C(\mathbf{e})$. The chromatic number of a disjoint union of cliques is simply the size of its largest clique:

$$\chi(G^n[T^n(\mathbf{P})]) = M_{\max} = \max_{\mathbf{e} \in T^n(\mathbf{P}_E)} |T^n(\mathbf{P}) \cap C(\mathbf{e})|.$$

By the lemma 2.6, the maximum number of typical sequences within a conditionally typical clique is bounded by $2^{nH(\mathbf{P}|e) + C_3\sqrt{n}}$. Hence:

$$\chi_\pi(n, \lambda) \leq \chi(G^n[T^n(\mathbf{P})]) = M_{\max} \leq 2^{nH(\mathbf{P}|e) + K_2\sqrt{n}}.$$

Setting $K = \max(K_1, K_2)$ simultaneously satisfies both the lower and upper bounds. Since K_1 and K_2 are independent of n and $\mathbf{P}$, the final constant K depends only on $\|X\|$ and λ , completing the proof. $\square$

Theorem 2.8. Let $G = (V, E)$ be a finite graph, and let P be a probability distribution on $V(G)$. Definitions 1.4, 1.5, and 1.7 of graph entropy are all equivalent. Specifically, the following quantities are identical:

$$H_1(G, P) = \lim_{t \rightarrow \infty} \min_{\substack{U \subseteq V^t \\ P^t(U) > 1-\varepsilon}} \frac{1}{t} \log \chi(G^t(U)),$$

$$H_2(G, P) = \min_{\substack{X \in \mathcal{S}(G) \\ P_X = P}} I(X; Y),$$

$$H_3(G, P) = \min_{\mathbf{a} \in VP(G)} - \sum_{i=1}^n p_i \log a_i,$$

where random variables X, Y take values in $V(G)$ with marginal P , and the set of stable sets $\mathcal{S}(G)$, respectively, and $0 < \varepsilon < 1$ is fix. That is,

$$H_1(G, P) = H_2(G, P) = H_3(G, P).$$

Proposition 2.9.

$$H_2(G, P) = H_3(G, P)$$

Proof. Let

$$V(G) = \{1, \dots, n\},$$

and let $\mathcal{S}(G)$ denote the family of maximal independent sets of G . Suppose that (X, Y) is a pair of random variables whose joint distribution attains the minimum in $H_2(G, P)$, with

$$P_X = P = (p_1, \dots, p_n).$$

Let Q denote the distribution of Y , i.e.,

$$Q(J) = \Pr(Y = J), \quad J \in \mathcal{S}(G),$$

and let

$$R(J | i) = \Pr(Y = J | X = i).$$

Since $\Pr(X \in Y) = 1$, we have

$$R(J | i) = 0 \quad \text{whenever} \quad i \notin J.$$

Moreover,

$$Q(J) = \sum_{i=1}^n \Pr(Y = J, X = i) = \sum_{i=1}^n P(X = i) \Pr(Y = J | X = i) = \sum_{i=1}^n p_i R(J | i).$$

Using the identity

$$I(X; Y) = H(Y) - H(Y | X),$$

and

$$H(Y | X) = E(H(Y | X = i)) \quad \text{where} \quad H(Y | X = i) = - \sum_{J \in \mathcal{S}(G)} R(J | i) \log_2 R(J | i),$$

we obtain

$$\begin{aligned} I(X; Y) &= - \sum_{J \in \mathcal{S}(G)} Q(J) \log_2 Q(J) + \sum_{i=1}^n p_i \sum_{J \in \mathcal{S}(G)} R(J | i) \log_2 R(J | i) \\ &= - \sum_{i=1}^n p_i \sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} R(J | i) \log_2 \frac{Q(J)}{R(J | i)}. \end{aligned}$$

Using Jensen's inequality and by the concavity of the logarithm,

$$\sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} R(J | i) \log_2 \frac{Q(J)}{R(J | i)} \leq \log_2 \left(\sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} R(J | i) \frac{Q(J)}{R(J | i)} \right) = \log_2 \left(\sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} Q(J) \right).$$

Consequently,

$$I(X; Y) \geq - \sum_{i=1}^n p_i \log_2 \left(\sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} Q(J) \right).$$

Define $\mathbf{a} = (a_1, \dots, a_n)$ by

$$a_i = \sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} Q(J), \quad i = 1, \dots, n.$$

Let $\mathbf{1}_J$ denote the characteristic vector of the kernel J . Then

$$\mathbf{a} = \sum_{J \in \mathcal{S}(G)} Q(J) \mathbf{1}_J.$$

Since Q is a probability distribution on $\mathcal{S}(G)$, and by Definition 1.2,

$$\mathbf{a} \in VP(G).$$

It follows that

$$I(X; Y) \geq - \sum_{i=1}^n p_i \log_2 a_i \geq \min_{\mathbf{a} \in VP(G)} \left\{ - \sum_{i=1}^n p_i \log_2 a_i \right\}.$$

Since (X, Y) was chosen to attain the minimum in $H_2(G, P)$, we conclude that

$$\min_{\substack{X \in \mathcal{Y} \in \mathcal{S}(G) \\ P_X = P}} I(X; Y) \geq \min_{\mathbf{a} \in VP(G)} \left\{ - \sum_{i=1}^n p_i \log_2 a_i \right\}.$$

Hence, $H_2(G, P) \geq H_3(G, P)$.

To prove the reverse inequality, let $\mathbf{a}$ be the minimizing vector in $H_3(G, P)$. Since $\mathbf{a} \in VP(G)$, it can be represented as

$$\mathbf{a} = \sum_{J \in \mathcal{S}(G)} Q'(J) \mathbf{1}_J \quad \text{where} \quad \sum_{J \in \mathcal{S}(G)} Q'(J) = 1.$$

Therefore,

$$a_i = \sum_{i \in J \in \mathcal{S}(G)} Q'(J),$$

and the vector of coefficients $Q'(J)$ defines a probability distribution on $\mathcal{S}(G)$.

Define $R'(J | i) = \frac{Q'(J)}{a_i}$ for $i \in J$, and $R'(J | i) = 0$ otherwise. We have

$$\sum_{J \in \mathcal{S}(G)} R'(J | i) = \sum_{i \in J \in \mathcal{S}(G)} \frac{Q'(J)}{a_i} = 1.$$

Therefore, $R'(J | i)$ defines a conditional probability distribution on $\mathcal{S}(G)$ given i . We further define another probability distribution Q^* on $\mathcal{S}(G)$ by

$$Q^*(J) = \sum_{i=1}^n p_i R'(J | i).$$

It defines a valid distribution since

$$\sum_{J \in \mathcal{S}(G)} Q^*(J) = \sum_{J \in \mathcal{S}(G)} \sum_{i=1}^n p_i R'(J | i) = \sum_{i=1}^n p_i \sum_{J \in \mathcal{S}(G)} R'(J | i) = \sum_{i=1}^n p_i = 1.$$

Considering the pair of random variables (X, Y) with $\text{dist}(X) = P$, $\text{dist}(Y) = Q^*$, and $\text{dist}(Y | X) = R'$. Following the same proof as in the previous part, we obtain

$$I(X; Y) = - \sum_{i=1}^n p_i \sum_{\substack{J \in \mathcal{S}(G) \\ i \in J}} R'(J | i) \log_2 \frac{Q^*(J)}{R'(J | i)}.$$

Therefore, the mutual information satisfies:

$$\min_{X \in Y \in \mathcal{S}(G), P_X = P} I(X; Y) \leq - \sum_{i=1}^n p_i \sum_{J \in \mathcal{S}(G)} R'(J | i) \log_2 \frac{Q^*(J)}{R'(J | i)}.$$

Using Jensen's inequality and the concavity of the logarithm, we have

$$\begin{aligned} \sum_{J \in \mathcal{S}(G)} Q^*(J) \log_2 \frac{Q'(J)}{Q^*(J)} &\leq \log_2 \left(\sum_{J \in \mathcal{S}(G)} Q^*(J) \frac{Q'(J)}{Q^*(J)} \right) = \log_2 \left(\sum_{J \in \mathcal{S}(G)} Q'(J) \right) = \log_2(1) = 0 \\ \implies \sum_{J \in \mathcal{S}(G)} Q^*(J) \log_2 \frac{Q'(J)}{Q^*(J)} &\leq 0. \end{aligned}$$

Now we have

$$\sum_{J \in \mathcal{S}(G)} Q^*(J) \log_2 \frac{Q'(J)}{Q^*(J)} = \sum_{J \in \mathcal{S}(G)} \sum_{i=1}^n p_i R'(J | i) [\log_2 Q'(J) - \log_2 Q^*(J)],$$

which implies

$$- \sum_{i,J} p_i R'(J | i) \log_2 Q^*(J) \leq - \sum_{i,J} p_i R'(J | i) \log_2 Q'(J).$$

Using this inequality, we obtain:

$$\begin{aligned} \min_{X \in Y \in \mathcal{S}(G), P_X = P} I(X; Y) &\leq - \sum_{i,J} p_i R'(J | i) \log_2 \frac{Q^*(J)}{R'(J | i)} \\ &= - \sum_{i,J} p_i R'(J | i) \log_2 Q^*(J) + \sum_{i,J} p_i R'(J | i) \log_2 R'(J | i) \\ &\leq - \sum_{i,J} p_i R'(J | i) \log_2 Q'(J) + \sum_{i,J} p_i R'(J | i) \log_2 R'(J | i) \\ &= - \sum_{i,J} p_i R'(J | i) \log_2 \frac{Q'(J)}{R'(J | i)}. \end{aligned}$$

Substituting $\frac{Q'(J)}{R'(J | i)} = a_i$ into the logarithm yields:

$$- \sum_{i,J} p_i R'(J | i) \log_2 \frac{Q'(J)}{R'(J | i)} = - \sum_{i=1}^n p_i \log_2 a_i \sum_{J \in \mathcal{S}(G)} R'(J | i) = - \sum_{i=1}^n p_i \log_2 a_i,$$

Since a was chosen to attain the minimum in $H_3(G, P)$, we conclude that

$$\min_{X \in Y \in \mathcal{S}(G), P_X = P} I(X; Y) \leq \min_{\mathbf{a} \in VP(G)} - \sum_{i=1}^n p_i \log_2 a_i,$$

which means $H_2(G, P) \leq H_3(G, P)$. □

Proposition 2.10.

$$H_2(G, P) \leq \liminf_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1-\lambda}} \frac{1}{n} \log_2 \chi(G^n(U))$$

Proof. Let $U \subseteq V(G^n)$ choose arbitrary with probability $P(U) \geq 1 - \lambda$. By Lemma 2.3, for any $\varepsilon > 0$ and sufficiently large n , the probability of the typical set satisfies $P(T^n(\mathbf{P})) \geq 1 - \varepsilon$. First, we establish a lower bound on the probability mass of the intersection $U \cap T^n(\mathbf{P})$:

$$P(U \cap T^n(\mathbf{P})) = P(U) + P(T^n(\mathbf{P})) - P(U \cup T^n(\mathbf{P})) \geq (1 - \lambda) + (1 - \varepsilon) - 1 = 1 - \varepsilon - \lambda. \quad (1)$$

Next, we estimate the chromatic number $\chi(U \cap T^n(\mathbf{P}))$. Let $\chi := \chi(U \cap T^n(\mathbf{P}))$, and partition $U \cap T^n(\mathbf{P})$ into χ independent chromatic classes $C_1, C_2, \dots, C_\chi \subseteq U \cap T^n(\mathbf{P})$. By extending each class C_j to a kernel $C \in S(G)^n \subseteq S(G^n)$, we have $C_j \subseteq C$, which implies $P(C_j) = P(C_j \cap T^n(\mathbf{P})) \leq P(C \cap T^n(\mathbf{P}))$. Consequently:

$$P(U \cap T^n(\mathbf{P})) = \sum_{j=1}^{\chi} P(C_j) \leq \chi(U \cap T^n(\mathbf{P})) \max_{\mathbf{A} \in S(G^n)} P(\mathbf{A} \cap T^n(\mathbf{P})). \quad (2)$$

To bound the maximum term, observe that for any $\mathbf{A} \in S(G^n)$:

$$P(\mathbf{A} \cap T^n(\mathbf{P})) = \sum_{\mathbf{x} \in \mathbf{A} \cap T^n(\mathbf{P})} P(\mathbf{x}) \leq \left(\max_{\mathbf{x} \in T^n(\mathbf{P})} P(\mathbf{x}) \right) |\mathbf{A} \cap T^n(\mathbf{P})|.$$

Taking the maximum over all $\mathbf{A} \in S(G^n)$ yields:

$$\max_{\mathbf{A} \in S(G^n)} P(\mathbf{A} \cap T^n(\mathbf{P})) \leq \left(\max_{\mathbf{x} \in T^n(\mathbf{P})} P(\mathbf{x}) \right) \cdot \left(\max_{\mathbf{A} \in S(G^n)} |\mathbf{A} \cap T^n(\mathbf{P})| \right). \quad (3)$$

Note that by Lemma 2.1, $\mathbf{A} \subseteq S(G)^n$. Let us consider an n -sequence $\mathbf{A} = (A_1, \dots, A_n)$ of kernels for which the value $|\mathbf{A} \cap T^n(\mathbf{P})|$ is maximum and a P -typical n -sequence $\mathbf{x} = (x_1, \dots, x_n)$ of the source contained in $\mathbf{A}$ (ie. $x_i \in A_i$ for all $1 \leq i \leq n$). Consider the joint sequence formed by aligning $\mathbf{x}$ and $\mathbf{A}$:

$$(\mathbf{x}, \mathbf{A}) = \begin{pmatrix} x_1 & x_2 & \dots & x_n \\ A_1 & A_2 & \dots & A_n \end{pmatrix}.$$

We use $N(y, B \mid \mathbf{x}, \mathbf{A})$ to count how many times the pair (y, B) appears in this double sequence, and $N(y \mid \mathbf{x})$ to count how many times the source letter y appears in $\mathbf{x}$.

We define a joint probability distribution Q over $V(G) \times S(G)$ for any pair (y, B) as follows:

$$Q(y, B) = \frac{N(y, B \mid \mathbf{x}, \mathbf{A})}{N(y \mid \mathbf{x})} \cdot P(y).$$

Here, the ratio $\frac{N(y, B \mid \mathbf{x}, \mathbf{A})}{N(y \mid \mathbf{x})}$ is simply the calculated conditional probability $Q(B \mid y)$, which makes sure that the total probability over $S(G)$ equals $P(y)$. To show that the joint sequence $(\mathbf{x}, \mathbf{A})$ is Q -typical, we calculate the absolute difference:

$$\begin{aligned} |N(y, B \mid \mathbf{x}, \mathbf{A}) - nQ(y, B)| &= \left| \frac{Q(y, B)}{P(y)} \cdot N(y \mid \mathbf{x}) - nQ(y, B) \right| \\ &= \frac{Q(y, B)}{P(y)} \cdot |N(y \mid \mathbf{x}) - nP(y)|. \end{aligned}$$

Since the source sequence $\mathbf{x}$ is P -typical, we have $|N(y \mid \mathbf{x}) - nP(y)| \leq K\sqrt{nP(y)}$ for a positive constant K . Plugging in this bound gives:

$$\begin{aligned} |N(y, B \mid \mathbf{x}, \mathbf{A}) - nQ(y, B)| &\leq \frac{Q(y, B)}{P(y)} \cdot K\sqrt{nP(y)} \\ &= K\sqrt{n \cdot \frac{Q^2(y, B)}{P(y)}} \leq K\sqrt{nQ(y, B)}, \end{aligned}$$

where the last step is true because $Q(y, B) \leq P(y)$. Thus, the sequence $(\mathbf{x}, \mathbf{A})$ is indeed Q -typical.

Next, we look at the marginal distribution $Q_{\mathbf{A}} = Q(B) = \sum_{y: y \in B} Q(y, B)$ over the set of kernels $S(G)$. The total count of kernel B in sequence $\mathbf{A}$ satisfies:

$$N(B \mid \mathbf{A}) - nQ(B) = \sum_{y \in B} (N(y, B \mid \mathbf{x}, \mathbf{A}) - nQ(y, B)).$$

Applying the triangle inequality and plugging in the Q -typicality bound of $(\mathbf{x}, \mathbf{A})$, we get:

$$\begin{aligned} |N(B | \mathbf{A}) - nQ(B)| &\leq \sum_{y \in V(G)} |N(y, B | \mathbf{x}, \mathbf{A}) - nQ(y, B)| \\ &\leq \sum_{y \in V(G)} K \sqrt{nQ(y, B)} = K \sqrt{n} \sum_{y \in V(G)} \sqrt{Q(y, B)}. \end{aligned}$$

To find an upper bound for the sum $\sum_{y \in V(G)} \sqrt{Q(y, B)}$, we use the Cauchy-Schwarz inequality $(\sum a_i b_i)^2 \leq (\sum a_i^2) (\sum b_i^2)$ with $a_y = 1$ and $b_y = \sqrt{Q(y, B)}$:

$$\sum_{y \in V(G)} \sqrt{Q(y, B)} \leq \sqrt{\sum_{y \in V(G)} 1^2} \cdot \sqrt{\sum_{y \in V(G)} Q(y, B)} = \sqrt{|V(G)|} \cdot \sqrt{Q(B)}.$$

Plugging this back in gives:

$$|N(B | \mathbf{A}) - nQ(B)| \leq K \sqrt{n} \left(\sqrt{|V(G)|} \sqrt{Q(B)} \right) = K_1 \sqrt{nQ(B)},$$

where $K_1 = K \sqrt{|V(G)|}$ is a constant that does not depend on n . This finishes the proof that the kernel sequence $\mathbf{A}$ is typical with respect to the marginal distribution $Q_{\mathbf{A}}$.

Consider the auxiliary graph Γ associated with the original probabilistic graph G , which define as

$$\begin{aligned} V(\Gamma) &= \{(x, A) : x \in V(G), A \in S(G), x \in A\}, \quad \text{and} \\ E(\Gamma) &= \{((x_1, A_1), (x_2, A_2)) : A_1 \neq A_2\}. \end{aligned}$$

By definition, $((x, A), (y, B)) \notin E(\Gamma) \iff A = B$, which induces an equivalence relation a on the vertices of Γ . Under this relation, two vertices (x, A) and (y, B) belong to the same equivalence class if and only if $A = B$.

We evaluate the complementary graph $\bar{\Gamma}$ equipped with the joint distribution Q over $V(G) \times S(G)$ and its marginal distribution $Q_{\mathbf{A}}$ defined over each $A \in S(G)$, forming the probabilistic graph $\bar{\Gamma} = [V(G) \times S(G), \bar{E}(\Gamma), Q]$. Since every equivalence class in $\bar{\Gamma}$ constitutes a complete subgraph (clique), Lemma 2.6 is applicable to $\bar{\Gamma}$.

Recall that $\mathbf{A}$ was shown to be a $Q_{\mathbf{A}}$ -typical sequence of kernels. Since each equivalence class corresponds to a kernel A with distribution $Q_{\mathbf{A}}$, the sequence $\mathbf{A} \in S(G)^n$ serves as a typical sequence of equivalence classes under the marginal distribution $Q_{\mathbf{A}}$. Consequently, by Lemma 2.6, the total number of Q -typical sequence pairs contained within the equivalence class sequence $\mathbf{A}$ satisfies:

$$2^{nH(Q|a) - K_2 \sqrt{n}} \leq |\mathbf{A} \cap T^n(Q)| \leq 2^{nH(Q|a) + K_2 \sqrt{n}},$$

where K_2 is a positive constant independent of n and the underlying distributions Q and $Q_{\mathbf{A}}$, depending solely on the alphabet size and graph structure.

To find an upper bound on the maximum number of typical source sequences inside any kernel sequence, we analyze the set of auxiliary distributions created from each valid sequence pair.

For any n -sequence of kernels $\mathbf{A} \in S(G)^n$ and any source sequence $\mathbf{x}' \in \mathbf{A} \cap T^n(P)$, the pair $(\mathbf{x}', \mathbf{A})$ is $Q_{\mathbf{x}'}$ -typical for a specific auxiliary distribution $Q_{\mathbf{x}'}$, defined similarly to Q :

$$Q_{\mathbf{x}'}(y, B) = \frac{N(y, B | \mathbf{x}', \mathbf{A})}{N(y | \mathbf{x}')} \cdot P(y).$$

For each arbitrary sequence $\mathbf{x}' \in \mathbf{A} \cap T^n(P)$, we count how many sequences $\mathbf{x} \in \mathbf{A}$ are, form a typical pair $(\mathbf{x}, \mathbf{A})$ under the induced distribution $Q_{\mathbf{x}'}$. Summing over all valid distributions $Q_{\mathbf{x}'}$ gives an upper bound on the total count:

$$\max_{\mathbf{A} \in S(G)^n} |\mathbf{A} \cap T^n(P)| \leq \sum_{Q_{\mathbf{x}'} : \mathbf{x}' \in \mathbf{A} \cap T^n(P)} |\{\mathbf{x} \in \mathbf{A} \mid (\mathbf{x}, \mathbf{A}) \text{ is } Q_{\mathbf{x}'}\text{-typical}\}|.$$

Since each pair $(y, B) \in V(G) \times S(G)$ can appear anywhere from 0 to n times in the n -sequence $(\mathbf{x}', \mathbf{A})$, there are at most $n+1$ possible counts for each pair. Thus, the total number of such auxiliary distributions $Q_{\mathbf{x}'}$ is bounded by:

$$(n+1)^{|V(G) \times S(G)|}.$$

Let $\mathcal{Q}$ be the set of all allowed auxiliary distributions over $V(G) \times S(G)$. Every distribution $Q_{\mathbf{x}'}$ belongs to $\mathcal{Q}$ since its marginal distribution over $V(G)$ keeps the original source distribution $P(y)$:

$$\sum_{B: y \in B} Q_{\mathbf{x}'}(y, B) = \frac{P(y)}{N(y \mid \mathbf{x}')} \sum_{B: y \in B} N(y, B \mid \mathbf{x}', \mathbf{A}) = P(y).$$

To limit the maximum number of typical sequences for any induced distribution $Q_{\mathbf{x}'}$, we extend the maximum across the entire set of allowed distributions $\mathcal{Q}$. Since $Q_{\mathbf{x}'} \in \mathcal{Q}$ for every valid sequence pair, the maximum number of typical pairs satisfies:

$$\begin{aligned} & \max_{Q_{\mathbf{x}'}: \mathbf{x}' \in \mathbf{A} \cap T^n(P)} |\{\mathbf{x} \in \mathbf{A} \mid (\mathbf{x}, \mathbf{A}) \text{ is } Q_{\mathbf{x}'}\text{-typical}\}| \\ & \leq \max_{Q' \in \mathcal{Q}} |\{\mathbf{x} \in \mathbf{A} \mid (\mathbf{x}, \mathbf{A}) \text{ is } Q'\text{-typical}\}| \\ & \leq 2^{n \cdot \max_{Q' \in \mathcal{Q}} H(Q'|a) + K_2 \sqrt{n}}. \end{aligned}$$

Bounding the sum by the total number of auxiliary distributions $Q_{\mathbf{x}'}$ multiplied by the maximum number of sequences $\mathbf{x} \in \mathbf{A}$ that form a typical pair $(\mathbf{x}, \mathbf{A})$ under any $Q_{\mathbf{x}'}$, we obtain:

$$\max_{\mathbf{A} \in S(G)^n} |\mathbf{A} \cap T^n(P)| \leq (n+1)^{|V(G) \times S(G)|} \cdot 2^{n \cdot \max_{Q' \in \mathcal{Q}} H(Q'|a) + K_2 \sqrt{n}}. \quad (4)$$

Here, K_2 depends only on the alphabet sizes and graph structure, remaining independent of both n and the specific choice of auxiliary distribution Q' . From statement (ii) of Lemma 2.3, the probability of any

sequence in the typical set $T^n(\mathbf{P})$ is bounded by

$$\max_{\mathbf{x} \in T^n(\mathbf{P})} P(\mathbf{x}) \leq 2^{-(nH(\mathbf{P}) - C\sqrt{n})}. \quad (5)$$

Combining (1), (2), and (3), we obtain

$$1 - \varepsilon - \lambda \leq P(U \cap T^n(\mathbf{P})) \leq \chi(U \cap T^n(\mathbf{P})) \cdot \max_{\mathbf{A} \in S(G)^n} |\mathbf{A} \cap T^n(\mathbf{P})| \cdot \max_{\mathbf{x} \in T^n(\mathbf{P})} P(\mathbf{x}).$$

Using (4) and (5) in the inequality yields

$$1 - \varepsilon - \lambda \leq \chi(U \cap T^n(\mathbf{P})) \cdot \left[(n+1)^{|V(G) \times S(G)|} \cdot 2^{n \max_{Q' \in \mathcal{Q}} H(Q'|a) + K_2 \sqrt{n}} \right] \cdot \left[2^{-nH(\mathbf{P}) + C\sqrt{n}} \right].$$

Expressing the polynomial factor as $(n+1)^{|V(G) \times S(G)|} = 2^{|V(G) \times S(G)| \log_2(n+1)}$ and setting $K = K_2 + C$, we consolidate all terms into a base-2 exponential expression:

$$1 - \varepsilon - \lambda \leq \chi(U \cap T^n(\mathbf{P})) \cdot \exp_2 \left(n \left(\max_{Q' \in \mathcal{Q}} H(Q'|a) - H(\mathbf{P}) \right) + K\sqrt{n} + |V(G) \times S(G)| \log_2(n+1) \right).$$

Finally, isolating $\chi(U \cap T^n(\mathbf{P}))$ gives

$$\chi(U \cap T^n(\mathbf{P})) \geq (1 - \varepsilon - \lambda) \cdot \exp_2 \left(n \left(H(\mathbf{P}) - \max_{Q' \in \mathcal{Q}} H(Q'|a) \right) - K\sqrt{n} - |V(G) \times S(G)| \log_2(n+1) \right).$$

Since $P(x)$ is the marginal distribution over $V(G)$, we have:

$$\begin{aligned} H(\mathbf{P}) - \max_{Q' \in \mathcal{Q}} H(Q'|a) &= \min_{Q' \in \mathcal{Q}} [H(\mathbf{P}) - H(Q'|a)] \\ &= \min_{X \in Y \subseteq S(G)} I(X; Y) = H_2(G, P). \end{aligned}$$

Notice that $\chi(U) \geq \chi(U \cap T^n(\mathbf{P}))$ naturally holds, because any valid coloring of the set E also works for its smaller part $U \cap T^n(\mathbf{P})$. Replacing the entropy term with $H_2(G, P)$ in our earlier result gives:

$$\chi(U) \geq (1 - \varepsilon - \lambda) \cdot \exp_2 \left(nH_2(G, P) - K\sqrt{n} - |V(G) \times S(G)| \log_2(n+1) \right),$$

for every set U where $P^n(U) \geq 1 - \lambda$.

Taking $\log_2$ of both sides, dividing by n , and picking the smallest possible value over all sets U with $P^n(U) \geq 1 - \lambda$, we get:

$$\frac{1}{n} \log_2 \left[\min_{P^n(U) \geq 1 - \lambda} \chi(U) \right] \geq \frac{1}{n} \log_2(1 - \varepsilon - \lambda) + H_2(G, P) - \frac{K}{\sqrt{n}} - \frac{|V(G) \times S(G)|}{n} \log_2(n + 1).$$

Letting $n \rightarrow \infty$, all the extra remainder terms drop to 0. This leaves us with the final lower bound:

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log_2 \min_{\substack{U \subseteq V^n \\ P^n(U) > 1 - \lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \geq H_2(G, P).$$

□

Proposition 2.11.

$$\limsup_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1 - \lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \leq H_2(G, P)$$

Proof. Let G be a finite simple graph with vertex set $V(G)$, and let P be a given probability distribution on $V(G)$. Let $S(G)$ denote the family of all maximal independent sets of G .

We fix a joint distribution $P_{X,Y}$ on $V(G) \times S(G)$ that achieves the minimum mutual information, so that $I(X; Y) = H_2(G, P)$ subject to $P_{X,Y}(x, Y) = 0$ whenever $x \notin Y$, and $\sum_{Y \in S(G)} P_{X,Y}(x, Y) = P(x)$ for all $x \in V(G)$.

The marginal distribution of Y on $S(G)$ induced by $P_{X,Y}$ is

$$P_Y(Y) = \sum_{x \in Y} P_{X,Y}(x, Y), \quad \forall Y \in S(G).$$

We extend P_Y to n -tuples of maximal independent sets $\mathbf{Y} = Y_1 \times Y_2 \times \cdots \times Y_n \in S(G)^n$ via the product distribution

$$P_{Y^n}(\mathbf{Y}) = \prod_{i=1}^n P_Y(Y_i).$$

Note that by Lemma 2.1, $\mathbf{Y}$ is a kernel in the co-normal power G^n of G .

To construct a covering of a high-probability subset of $V(G)^n$, we select K sequence-independent sets $\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_K \in S(G)^n$ independently and identically distributed according to P_{Y^n} . The joint distribution over the choice of these K sets is

$$P_K[\mathbf{Y}_1, \dots, \mathbf{Y}_K] = \prod_{j=1}^K P_{Y^n}(\mathbf{Y}_j).$$

Let $U[\mathbf{Y}_1, \dots, \mathbf{Y}_K] = \bigcup_{j=1}^K \mathbf{Y}_j \subseteq V(G)^n$ denote the set of sequences covered by the randomly selected $\mathbf{Y}_1, \dots, \mathbf{Y}_K$. The expected probability mass of the uncovered set $U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K]$ under P_K is

$$\begin{aligned} \mathbb{E}_{P_K} [P^n(U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K])] &= \sum_{\mathbf{Y}_1, \dots, \mathbf{Y}_K} P_K[\mathbf{Y}_1, \dots, \mathbf{Y}_K] P^n(U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K]) \\ &= \sum_{\mathbf{Y}_1, \dots, \mathbf{Y}_K} P_K[\mathbf{Y}_1, \dots, \mathbf{Y}_K] \sum_{\substack{\mathbf{x} \in V(G)^n \\ \mathbf{x} \notin \bigcup_{j=1}^K \mathbf{Y}_j}} P^n(\mathbf{x}) \\ &= \sum_{\mathbf{Y}_1, \dots, \mathbf{Y}_K} P_K[\mathbf{Y}_1, \dots, \mathbf{Y}_K] \sum_{\mathbf{x} \in V(G)^n} P^n(\mathbf{x}) \cdot \mathbb{I}_{\mathbf{x} \notin \bigcup_{j=1}^K \mathbf{Y}_j} \\ &= \sum_{\mathbf{x} \in V(G)^n} P^n(\mathbf{x}) \sum_{\mathbf{Y}_1, \dots, \mathbf{Y}_K} P_K[\mathbf{Y}_1, \dots, \mathbf{Y}_K] \cdot \mathbb{I}_{\mathbf{x} \notin \bigcup_{j=1}^K \mathbf{Y}_j} \\ &= \sum_{\mathbf{x} \in V(G)^n} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}), \end{aligned} \tag{1}$$

where $\mathbb{I}_{\{\cdot\}}$ denotes the indicator function, and $\Pr_{P_K}(\mathbf{x} \text{ is not covered})$ is the probability that a specific sequence $\mathbf{x} \in V(G)^n$ is not contained in any of $\mathbf{Y}_1, \dots, \mathbf{Y}_K$.

We partition $V(G)^n$ into typical sequences $T^n(P)$ and non-typical sequences $\overline{T^n(P)}$. By Lemma 2.3 (i), for any $\lambda > 0$, there exists a constant C_1 such that

$$P^n(\overline{T^n(P)}) < \frac{\lambda}{2}. \quad (2)$$

For any typical sequence $\mathbf{x} \in T^n(P)$, a sequence $\mathbf{Y} \in S(G)^n$ covers $\mathbf{x}$ if $\mathbf{x} \in \mathbf{Y}$. Since the K sequence-independent sets are drawn independently,

$$\Pr_{P_K}(\mathbf{x} \text{ is not covered}) = \left[1 - \sum_{\mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K. \quad (3)$$

To bound the number of typical sequences $\mathbf{Y} \in S(G)^n$ that cover $\mathbf{x}$, We construct an auxiliary graph $\tilde{\Gamma}$ on $V(G) \times S(G)$ such that

$$\begin{aligned} V(\tilde{\Gamma}) &= \{(x, A) : x \in V(G), A \in S(G), x \in A\}, \quad \text{and} \\ E(\tilde{\Gamma}) &= \{((x_1, A_1), (x_2, A_2)) : x_1 = x_2\} \end{aligned}$$

Now, define the equivalence relation e on $V(\tilde{\Gamma})$ by $(x_1, A_1) e (x_2, A_2) \iff x_1 = x_2$. By construction, the equivalence classes $e(x) = \{(x, A) : A \in S(G), x \in A\}$ partition $V(\tilde{\Gamma})$ into pairwise disjoint complete subgraphs (cliques). Thus, $\tilde{\Gamma}$ strictly satisfies the hypotheses of Lemma 2.6.

We then consider the graph $\tilde{\Gamma}^n$, whose vertices are sequence pairs $(\mathbf{x}, \mathbf{Y}) \in V(G)^n \times S(G)^n$. The equivalence relation e^n on $V(\tilde{\Gamma}^n)$ is induced componentwise:

$$(\mathbf{x}_1, \mathbf{Y}_1) e^n (\mathbf{x}_2, \mathbf{Y}_2) \iff \mathbf{x}_1 = \mathbf{x}_2.$$

Since e^n inherits reflexivity, symmetry, and transitivity componentwise, e^n remains a valid equivalence relation on $V(\tilde{\Gamma}^n)$. Consequently, for a sequence of vertices $\mathbf{x} = (x_1, \dots, x_n) \in V(G)^n$, the equivalence class

$$e^n(\mathbf{x}) = e(x_1) \times e(x_2) \times \dots \times e(x_n)$$

forms a pairwise disjoint clique in $\tilde{\Gamma}^n$, preserving the exact structure required by Lemma 2.6.

For a fixed typical sequence of vertices $\mathbf{x} = (x_1, \dots, x_n) \in T^n(P_X)$, the corresponding product equivalence class $e^n(\mathbf{x}) = e(x_1) \times \dots \times e(x_n)$ in $\tilde{\Gamma}^n$ consists of all sequence pairs $(\mathbf{x}, \mathbf{Y}) \in V(G)^n \times S(G)^n$ such that $x_i \in Y_i$ for all $1 \leq i \leq n$.

For each $(\mathbf{x}, \mathbf{Y}) \in T^n(P_{X,Y})$, completely similar to the proof of the kernel sequence $\mathbf{A}$ being typical with respect to the marginal distribution $Q_{\mathbf{A}}$ in Proposition 2.10, we have $\mathbf{Y} \in T^n(P_{Y^n})$.

As a result, the number of typical sequences $(\mathbf{x}, \mathbf{Y})$ belonging to $e^n(\mathbf{x})$ is at most equivalent to counting the number of typical sequences of maximal stable sets $\mathbf{Y} = (Y_1, \dots, Y_n) \in S(G)^n$ that cover $\mathbf{x}$, denoted by $N(\mathbf{x})$.

Note that, given the structure of the equivalence classes in the graph, for every i ,

$$P(e(x_i)) = \sum_{(x_i, Y); x_i \in Y} P_{X,Y}(x_i, Y) = P(x_i)$$

so, the distribution $P(e^n)$ for $e^n(\mathbf{x}) = (e(x_1), \dots, e(x_n))$ is the same as P on $V(G)$. Since $\mathbf{x}$ is typical, $e^n(\mathbf{x})$ is also a P -typical sequence.

Now, by Lemma 2.6, the number $N(\mathbf{x})$ of such covering sequences $\mathbf{Y} \in S(G)^n$ satisfies the following bound:

$$2^{nH(P_{X,Y|e}) - C_2\sqrt{n}} \leq |T^n(P_{X,Y}) \cap e^n(\mathbf{x})| \leq N(\mathbf{x}),$$

where $C_2 > 0$ is a numerical constant independent of n .

Furthermore, by Lemma 2.3, each such $P_{X,Y}$ -typical sequence $\mathbf{Y}$ satisfies

$$P_{Y^n}(\mathbf{Y}) \geq 2^{-nH(P_Y) - C_3\sqrt{n}},$$

where $C_3 > 0$. Summing $P_{Y^n}(\mathbf{Y})$ over all such typical $\mathbf{Y}$ yields

$$\begin{aligned} \sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) &\geq N(\mathbf{x}) \cdot 2^{-nH(P_Y) - C_3\sqrt{n}} \geq 2^{nH(P_{X,Y|e}) - C_2\sqrt{n}} \cdot 2^{-nH(P_Y) - C_3\sqrt{n}} \\ &= 2^{-n(H(P_Y) - H(P_{X,Y|e})) - C_4\sqrt{n}} \\ &= 2^{-nI(X;Y) - C_4\sqrt{n}} \\ &= 2^{-nH_2(G,P) - C_4\sqrt{n}}, \end{aligned}$$

where $C_4 = C_2 + C_3$, using $H_2(G, P) = I(X; Y) = H(Y) - H(Y | X) = H(P_Y) - H(P_{X,Y} | e)$. Using the inequality $(1 - t)^K \leq \exp_2(-Kt)$, we have

$$\left[1 - \sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K \leq \exp_2 \left(-K \sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right).$$

Choose the number of maximal stable sets to be

$$K = \left\lfloor 2^{n(H_2(G, P) + \delta)} \right\rfloor,$$

and using the lower bound on $\sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y})$ yields

$$\begin{aligned} \left[1 - \sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K &\leq \exp_2 \left(-2^{n(H_2(G, P) + \delta)} \cdot 2^{-nH_2(G, P) - C_4\sqrt{n}} \right) \\ &= \exp_2 \left(-2^{n\delta - C_4\sqrt{n}} \right). \end{aligned}$$

For any fixed $\delta > 0$, $n\delta - C_4\sqrt{n} \rightarrow \infty$ as $n \rightarrow \infty$. Consequently, $\exp_2 \left(-2^{n\delta - C_4\sqrt{n}} \right) \rightarrow 0$. Thus, for sufficiently large n ,

$$\left[1 - \sum_{\mathbf{x} \in \mathbf{Y} \in S(G)^n} P_{Y^n}(\mathbf{Y}) \right]^K < \frac{\lambda}{2}, \quad \forall \mathbf{x} \in T^n(P). \quad (4)$$

From (1), (2), (3), and (4), bounding the expected probability mass of uncovered vertices across all $V(G)^n$ yields:

$$\begin{aligned} \mathbb{E}_{P_K} [P^n(U^c[\mathbf{Y}_1, \dots, \mathbf{Y}_K])] &= \sum_{\mathbf{x} \in V(G)^n} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) \\ &= \sum_{\mathbf{x} \in T^n(P)} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) + \sum_{\mathbf{x} \in \overline{T^n(P)}} P^n(\mathbf{x}) \Pr_{P_K}(\mathbf{x} \text{ is not covered}) \\ &< \frac{\lambda}{2} \sum_{\mathbf{x} \in T^n(P)} P^n(\mathbf{x}) + \sum_{\mathbf{x} \in \overline{T^n(P)}} P^n(\mathbf{x}) \cdot 1 \\ &< \frac{\lambda}{2} \cdot 1 + P^n(\overline{T^n(P)}) < \frac{\lambda}{2} + \frac{\lambda}{2} = \lambda, \end{aligned}$$

for sufficiently large n .

Since the expectation over all choices of $(\mathbf{Y}_1, \dots, \mathbf{Y}_K)$ is strictly less than λ , there exists at least one specific realization $(\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*)$ such that

$$P^n(U^c[\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*]) < \lambda \implies P^n(U[\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*]) \geq 1 - \lambda.$$

Let $T_n = U[\mathbf{Y}_1^*, \dots, \mathbf{Y}_K^*] \subseteq V(G)^n$. Since each $\mathbf{Y}_j^*$ is a maximal independent set in G^n , the induced subgraph $G^n(T_n)$ is covered by K kernels. Therefore, by Lemma 2.2 $G^n(T_n)$ can be properly colored using at most K colors:

$$\chi(G^n(T_n)) \leq K = \left\lfloor 2^{n(H_2(G, P) + \delta)} \right\rfloor.$$

Taking the logarithm and dividing by n , we have

$$\min_{\substack{U \subseteq V^n \\ P^t(U) > 1 - \lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \leq \frac{1}{n} \log_2 \chi(G^n(T_n)) \leq H_2(G, P) + \delta.$$

Evaluating the limit superior as $n \rightarrow \infty$:

$$\limsup_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1 - \lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \leq H_2(G, P) + \delta.$$

Since $\delta > 0$ was arbitrary, this completes the proof. $\square$

Proof of Theorem 2.8. By Proposition 2.9, we establish the equivalence of $H_2(G, P)$ and $H_3(G, P)$. Combining Proposition 2.11 and Proposition 2.10, we obtain:

$$\limsup_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1-\lambda}} \frac{1}{n} \log_2 \chi(G^n(U)) \leq H_2(G, P) \leq \liminf_{n \rightarrow \infty} \min_{\substack{U \subseteq V^n \\ P^n(U) > 1-\lambda}} \frac{1}{n} \log_2 \chi(G^n(U)).$$

For brevity, let $\chi(n, \lambda) := \min_{\substack{U \subseteq V^n \\ P^n(U) > 1-\lambda}} \chi(G^n(U))$.

Since the inequality $\liminf_{n \rightarrow \infty} \frac{1}{n} \log_2 \chi(n, \lambda) \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \log_2 \chi(n, \lambda)$ always holds, it follows that:

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log_2 \chi(n, \lambda) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log_2 \chi(n, \lambda) = H_2(G, P).$$

This implies that $\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \chi(n, \lambda)$ exists and is equal to $H_2(G, P)$. By definition, this directly yields $H_1(G, P) = H_2(G, P)$.

Altogether, we conclude that:

$$H_1(G, P) = H_2(G, P) = H_3(G, P),$$

which completes the proof. □

3 Maximum Graph Entropy and Fractional Chromatic Number

Lemma 3.1. *Let $C \subseteq \mathbb{R}^n$ be a nonempty, closed, and convex set.*

$$\mathcal{P}_C(\mathbf{x}) := \arg \min_{\mathbf{y} \in C} \|\mathbf{x} - \mathbf{y}\|.$$

Then:

1. *For any $\mathbf{x} \in \mathbb{R}^n$, the projection $\mathcal{P}_C(\mathbf{x})$ is well-defined (exists and is unique).*
2. *Given $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} = \mathcal{P}_C(\mathbf{x})$, for any $\mathbf{z} \in C$, it holds that*

$$\langle \mathbf{x} - \mathbf{y}, \mathbf{z} - \mathbf{y} \rangle \leq 0.$$

Proof. Existence: Since $C \neq \emptyset$, choose any $\mathbf{z}_0 \in C$ and let $r = \|\mathbf{x} - \mathbf{z}_0\|$. The intersection $\overline{\mathcal{B}(\mathbf{x}, r)} \cap C$ is nonempty, bounded, and closed in $\mathbb{R}^n$, hence compact. The continuous distance function $g(\mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$ achieves its minimum on this compact set by the Extreme Value Theorem.

Uniqueness: Suppose there exist $\mathbf{y}_1, \mathbf{y}_2 \in C$ with $\mathbf{y}_1 \neq \mathbf{y}_2$ such that $\text{dist}(\mathbf{x}, C) = \|\mathbf{x} - \mathbf{y}_1\| = \|\mathbf{x} - \mathbf{y}_2\| = d$. Let $\mathbf{y}_c = \frac{1}{2}(\mathbf{y}_1 + \mathbf{y}_2)$. Since C is convex, $\mathbf{y}_c \in C$, and thus $\|\mathbf{x} - \mathbf{y}_c\| \geq d$. However, expanding the norm yields:

$$\begin{aligned} \|\mathbf{x} - \mathbf{y}_c\|^2 &= \left\| \mathbf{x} - \frac{\mathbf{y}_1 + \mathbf{y}_2}{2} \right\|^2 = \left\| \frac{1}{2}(\mathbf{x} - \mathbf{y}_1) + \frac{1}{2}(\mathbf{x} - \mathbf{y}_2) \right\|^2 \\ &= \frac{1}{4} \|(\mathbf{x} - \mathbf{y}_1) + (\mathbf{x} - \mathbf{y}_2)\|^2 \\ &= \frac{1}{4} (\|\mathbf{x} - \mathbf{y}_1\|^2 + \|\mathbf{x} - \mathbf{y}_2\|^2 + 2\langle \mathbf{x} - \mathbf{y}_1, \mathbf{x} - \mathbf{y}_2 \rangle) \\ &= \frac{1}{4} (\|\mathbf{x} - \mathbf{y}_1\|^2 + \|\mathbf{x} - \mathbf{y}_2\|^2 + \|\mathbf{x} - \mathbf{y}_1\|^2 + \|\mathbf{x} - \mathbf{y}_2\|^2 - \|(\mathbf{x} - \mathbf{y}_1) - (\mathbf{x} - \mathbf{y}_2)\|^2) \\ &= d^2 - \frac{1}{4} \|\mathbf{y}_1 - \mathbf{y}_2\|^2 \end{aligned}$$

Subtracting d^2 gives:

$$0 \leq \|\mathbf{x} - \mathbf{y}_c\|^2 - d^2 = -\frac{1}{4} \|\mathbf{y}_1 - \mathbf{y}_2\|^2 < 0,$$

which is a contradiction. Hence, the minimizer is unique.

Variational Inequality: For any $\mathbf{z} \in C$ and $t \in (0, 1)$, by convexity $\mathbf{y} + t(\mathbf{z} - \mathbf{y}) \in C$. Since $\mathbf{y} = \mathcal{P}_C(\mathbf{x})$ minimizes distance over C :

$$\|\mathbf{x} - (\mathbf{y} + t(\mathbf{z} - \mathbf{y}))\|^2 \geq \|\mathbf{x} - \mathbf{y}\|^2.$$

Expanding the left hand side:

$$\|\mathbf{x} - \mathbf{y}\|^2 - 2t\langle \mathbf{x} - \mathbf{y}, \mathbf{z} - \mathbf{y} \rangle + t^2\|\mathbf{z} - \mathbf{y}\|^2 \geq \|\mathbf{x} - \mathbf{y}\|^2.$$

Canceling $\|\mathbf{x} - \mathbf{y}\|^2$ and dividing by $t > 0$:

$$-2\langle \mathbf{x} - \mathbf{y}, \mathbf{z} - \mathbf{y} \rangle + t\|\mathbf{z} - \mathbf{y}\|^2 \geq 0.$$

Taking the limit as $t \rightarrow 0^+$ yields $\langle \mathbf{x} - \mathbf{y}, \mathbf{z} - \mathbf{y} \rangle \leq 0$. □

Corollary 3.2. *Let $C \subseteq \mathbb{R}^n$ be a nonempty, closed, and convex set. For any $\mathbf{x} \notin C$, there exists $\mathbf{w} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that*

$$\sup_{\mathbf{z} \in C} \langle \mathbf{w}, \mathbf{z} \rangle < \langle \mathbf{w}, \mathbf{x} \rangle.$$

Proof. Let $\mathbf{y} = \mathcal{P}_C(\mathbf{x})$ and define $\mathbf{w} = \mathbf{x} - \mathbf{y}$. Since $\mathbf{x} \notin C$, $\mathbf{w} \neq \mathbf{0}$. By Lemma 3.1, for any $\mathbf{z} \in C$:

$$\langle \mathbf{w}, \mathbf{z} - \mathbf{y} \rangle \leq 0 \implies \langle \mathbf{w}, \mathbf{z} \rangle \leq \langle \mathbf{w}, \mathbf{y} \rangle = \langle \mathbf{w}, \mathbf{x} \rangle - \|\mathbf{w}\|^2.$$

Taking the supremum over all $\mathbf{z} \in C$:

$$\sup_{\mathbf{z} \in C} \langle \mathbf{w}, \mathbf{z} \rangle \leq \langle \mathbf{w}, \mathbf{x} \rangle - \|\mathbf{w}\|^2 < \langle \mathbf{w}, \mathbf{x} \rangle,$$

since $\|\mathbf{w}\|^2 > 0$. □

Theorem 3.3 (Supporting Hyperplane Theorem). *Given a nonempty convex set $C \subseteq \mathbb{R}^n$ and a point $\mathbf{x}_0 \in \partial C$, there exists $\mathbf{w} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that*

$$\forall \mathbf{z} \in C, \quad \langle \mathbf{w}, \mathbf{z} \rangle \leq \langle \mathbf{w}, \mathbf{x}_0 \rangle.$$

That is, $P = \{\mathbf{z} \in \mathbb{R}^n \mid \mathbf{w}^T \mathbf{z} = \mathbf{w}^T \mathbf{x}_0\}$ is a supporting hyperplane of C at $\mathbf{x}_0$.

Proof. We consider two cases depending on the interior of C :

Case 1: $\text{int}(C) = \emptyset$.

If $\text{int}(C) = \emptyset$, then C lies in an affine set of dimension strictly less than n . Thus, there exists a hyperplane containing C entirely, say $H = \{\mathbf{z} \in \mathbb{R}^n \mid \langle \mathbf{w}, \mathbf{z} \rangle = b\}$ with $\mathbf{w} \neq \mathbf{0}$. Since $\mathbf{x}_0 \in \partial C \subseteq \text{cl}(C) \subseteq H$, we have $b = \langle \mathbf{w}, \mathbf{x}_0 \rangle$. Hence $\langle \mathbf{w}, \mathbf{z} \rangle = \langle \mathbf{w}, \mathbf{x}_0 \rangle$ for all $\mathbf{z} \in C$, completing the proof.

Case 2: $\text{int}(C) \neq \emptyset$.

For each $k \in \mathbb{N}$, define the set

$$C_k := \left\{ \mathbf{z} \in \mathbb{R}^n \mid \mathcal{B}\left(\mathbf{z}, \frac{1}{k}\right) \subseteq \text{cl}(C) \right\}.$$

Note that C_k is closed, convex, and non-empty for sufficiently large k . Since $\mathbf{x}_0 \in \partial C$, $\mathbf{x}_0 \notin \text{int}(C)$, which implies $\mathbf{x}_0 \notin C_k$ for all k since $\text{int}(C) = \text{int}(\text{cl}(C))$ by convexity.

By Corollary 3.2, for each k , there exists a vector $\mathbf{w}_k \neq \mathbf{0}$ strictly separating C_k and $\mathbf{x}_0$:

$$\forall \mathbf{z} \in C_k, \quad \langle \mathbf{w}_k, \mathbf{z} \rangle < \langle \mathbf{w}_k, \mathbf{x}_0 \rangle.$$

Without loss of generality, normalize $\mathbf{w}_k$ so that $\|\mathbf{w}_k\| = 1$.

The sequence $\{\mathbf{w}_k\}_{k=1}^\infty$ lies on the compact unit sphere $\mathbb{S}^{n-1}$. By the Bolzano–Weierstrass Theorem, there exists a convergent subsequence $\{\mathbf{w}_{k_j}\}_{j=1}^\infty$ converging to a limit $\mathbf{w}$ with $\|\mathbf{w}\| = 1$ (hence $\mathbf{w} \neq \mathbf{0}$).

For any fixed $\mathbf{x} \in \text{int}(C)$, there exists $\varepsilon > 0$ such that $\mathcal{B}(\mathbf{x}, \varepsilon) \subseteq C$. Choosing N such that $1/N < \varepsilon$, we have $\mathbf{x} \in C_k$ for all $k > N$. Therefore:

$$\langle \mathbf{w}_{k_j}, \mathbf{x} \rangle < \langle \mathbf{w}_{k_j}, \mathbf{x}_0 \rangle \quad \text{for all } k_j > N.$$

Taking the limit as $j \rightarrow \infty$ on both sides yields:

$$\langle \mathbf{w}, \mathbf{x} \rangle \leq \langle \mathbf{w}, \mathbf{x}_0 \rangle \quad \forall \mathbf{x} \in \text{int}(C).$$

Finally, for any $\mathbf{y} \in C$, by the line segment principle for convex sets with non-empty interior, there exists a sequence $\{\mathbf{y}_m\}_{m=1}^\infty \subset \text{int}(C)$ such that $\mathbf{y}_m \rightarrow \mathbf{y}$ as $m \rightarrow \infty$. Since $\langle \mathbf{w}, \mathbf{y}_m \rangle \leq \langle \mathbf{w}, \mathbf{x}_0 \rangle$ for all m , taking $m \rightarrow \infty$ gives:

$$\langle \mathbf{w}, \mathbf{y} \rangle \leq \langle \mathbf{w}, \mathbf{x}_0 \rangle \quad \forall \mathbf{y} \in C.$$

This completes the proof. $\square$

Corollary 3.4. *If $\mathcal{A} \subset \mathbb{R}_+^n$ is a convex corner and $\mathbf{a} \in \mathcal{A}$ is a maximal vector (i.e., no $\mathbf{a}' \in \mathcal{A}$ satisfies $\mathbf{a}' \geq \mathbf{a}$ with $\mathbf{a}' \neq \mathbf{a}$), then there exist a non-negative vector $\mathbf{b} \in \mathbb{R}_+^n$ normalized such that:*

$$\mathbf{b}^T \mathbf{x} \leq 1 \quad \forall \mathbf{x} \in \mathcal{A}, \quad \text{and} \quad \mathbf{b}^T \mathbf{a} = 1.$$

Proof. Let $\mathcal{A} \subset \mathbb{R}_+^n$ be a convex corner and let $\mathbf{a} \in \mathcal{A}$ be a maximal element of $\mathcal{A}$.

Consider the closed shifted positive orthant $\bar{\mathcal{B}} = \{\mathbf{y} \in \mathbb{R}^n \mid \mathbf{y} \geq \mathbf{a}\}$, which is a convex set. Define the set difference:

$$\mathcal{C} = \mathcal{A} - \bar{\mathcal{B}} = \{\mathbf{x} - \mathbf{y} \mid \mathbf{x} \in \mathcal{A}, \mathbf{y} \in \bar{\mathcal{B}}\}.$$

Since $\mathcal{A}$ and $\bar{\mathcal{B}}$ are convex sets, their Minkowski difference $\mathcal{C}$ is also convex and non-empty. Note that $\mathbf{0} = \mathbf{a} - \mathbf{a} \in \mathcal{C}$. Furthermore, since $\mathbf{a}$ is maximal in $\mathcal{A}$, no element $\mathbf{x} \in \mathcal{A}$ satisfies $\mathbf{x} \gg \mathbf{a}$. This implies that $\mathcal{C} \cap \{\mathbf{z} \in \mathbb{R}^n \mid \mathbf{z} \gg \mathbf{0}\} = \emptyset$. Consequently, the origin $\mathbf{0}$ cannot be an interior point of $\mathcal{C}$, so $\mathbf{0} \in \partial \mathcal{C}$.

By the Supporting Hyperplane Theorem 3.3 applied to the convex set $\mathcal{C}$ at the boundary point $\mathbf{0}$, there exists a non-zero vector $\mathbf{b}_0 \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that:

$$\mathbf{b}_0^T \mathbf{z} \leq \mathbf{b}_0^T \mathbf{0} = 0, \quad \forall \mathbf{z} \in \mathcal{C}.$$

Substituting $\mathbf{z} = \mathbf{x} - \mathbf{y}$ for any $\mathbf{x} \in \mathcal{A}$ and $\mathbf{y} \in \overline{\mathcal{B}}$, we obtain $\mathbf{b}_0^T(\mathbf{x} - \mathbf{y}) \leq 0$, or equivalently:

$$\mathbf{b}_0^T \mathbf{x} \leq \mathbf{b}_0^T \mathbf{y}, \quad \forall \mathbf{x} \in \mathcal{A}, \forall \mathbf{y} \in \overline{\mathcal{B}}.$$

Since $\mathbf{a} \in \mathcal{A}$ and $\mathbf{a} \in \overline{\mathcal{B}}$, setting $\mathbf{y} = \mathbf{a}$ yields $\mathbf{b}_0^T \mathbf{x} \leq \mathbf{b}_0^T \mathbf{a}$ for all $\mathbf{x} \in \mathcal{A}$, and setting $\mathbf{x} = \mathbf{a}$ yields $\mathbf{b}_0^T \mathbf{a} \leq \mathbf{b}_0^T \mathbf{y}$ for all $\mathbf{y} \in \overline{\mathcal{B}}$. Combining these gives:

$$\mathbf{b}_0^T \mathbf{x} \leq \mathbf{b}_0^T \mathbf{a} \leq \mathbf{b}_0^T \mathbf{y}, \quad \forall \mathbf{x} \in \mathcal{A}, \forall \mathbf{y} \in \overline{\mathcal{B}}. \quad (2)$$

First, we show that $\mathbf{b}_0 \geq \mathbf{0}$. Suppose for contradiction that there exists some index $i \in \{1, \dots, n\}$ such that $(\mathbf{b}_0)_i < 0$. For any $\epsilon > 0$, consider the vector $\mathbf{y}_\epsilon = \mathbf{a} + \epsilon \mathbf{e}_i$, where $\mathbf{e}_i$ is the i -th standard basis vector. Since $\mathbf{y}_\epsilon \geq \mathbf{a}$, we have $\mathbf{y}_\epsilon \in \overline{\mathcal{B}}$. Substituting $\mathbf{y}_\epsilon$ into the right-hand inequality of (2) yields:

$$\mathbf{b}_0^T \mathbf{a} \leq \mathbf{b}_0^T (\mathbf{a} + \epsilon \mathbf{e}_i) = \mathbf{b}_0^T \mathbf{a} + \epsilon (\mathbf{b}_0)_i.$$

Subtracting $\mathbf{b}_0^T \mathbf{a}$ from both sides gives $\epsilon (\mathbf{b}_0)_i \geq 0$, which implies $(\mathbf{b}_0)_i \geq 0$ since $\epsilon > 0$. This contradicts the assumption that $(\mathbf{b}_0)_i < 0$. Thus, $\mathbf{b}_0 \in \mathbb{R}_+^n$.

Next, since $\mathbf{a} \in \mathcal{A} \subset \mathbb{R}_+^n$ is maximal, $\mathbf{a} > \mathbf{0}$. Given that $\mathbf{b}_0 \geq \mathbf{0}$ and $\mathbf{b}_0 \neq \mathbf{0}$, we have $\mathbf{b}_0^T \mathbf{a} > 0$.

Now, define the normalized vector:

$$\mathbf{b} = \frac{\mathbf{b}_0}{\mathbf{b}_0^T \mathbf{a}}.$$

Since $\mathbf{b}_0 \geq \mathbf{0}$ and $\mathbf{b}_0^T \mathbf{a} > 0$, it immediately follows that $\mathbf{b} \in \mathbb{R}_+^n$.

Finally, multiplying inequality (2) by the positive scalar $\frac{1}{\mathbf{b}_0^T \mathbf{a}}$, we obtain:

$$\mathbf{b}^T \mathbf{x} \leq 1 \quad \forall \mathbf{x} \in \mathcal{A},$$

and by construction:

$$\mathbf{b}^T \mathbf{a} = \frac{\mathbf{b}_0^T \mathbf{a}}{\mathbf{b}_0^T \mathbf{a}} = 1,$$

which completes the proof. $\square$

Lemma 3.5. *Suppose $a \in VP(G)$ is a maximal vector in $VP(G)$, i.e., there exists no vector $a' \in VP(G) \setminus \{a\}$ such that $a' \geq a$ coordinatewise. Then, there exists a probability distribution P on $V(G)$ such that the value of $H(G, P)$ is attained at a .*

Proof. Recall that $A = VP(G)$ is a convex corner in $\mathbb{R}_+^k$ and satisfies the down-monotonicity property: if $x \in A$ and $0 \leq x' \leq x$ coordinatewise, then $x' \in A$.

Since $a \in A$ is a maximal element, it lies on the upper boundary $\partial^+ A$. By Corollary 3.4, there exists a non-zero vector $b = (b_1, \dots, b_k) \in \mathbb{R}_+^k$ such that:

$$b^T x \leq 1 \quad \forall x \in A, \quad \text{and} \quad b^T a = \sum_{i=1}^k b_i a_i = 1.$$

Define the vector $p = (p_1, \dots, p_k)$ as the coordinatewise product of a and b :

$$p_i = a_i \cdot b_i, \quad \text{or equivalently } p = a \circ b.$$

Since $a_i \geq 0$ and $b_i \geq 0$ for all i , we have $p_i \geq 0$. Also, the sum of the coordinates satisfies:

$$\sum_{i=1}^k p_i = \sum_{i=1}^k a_i b_i = b^T a = 1.$$

Hence, p defines a valid probability distribution P on $V(G)$. By Definition 1.4, the graph entropy with respect to P is given by:

$$H(G, P) = H_{VP(G)}(p) = \min_{x \in VP(G)} \left(- \sum_{i=1}^k p_i \log_2 x_i \right).$$

Let $x = (x_1, \dots, x_k)$ be an arbitrary vector in $VP(G)$. Applying Jensen's inequality to the strictly concave logarithm function, we obtain:

$$\sum_{i=1}^k p_i \log_2 \left(\frac{x_i}{a_i} \right) \leq \log_2 \left(\sum_{i=1}^k p_i \frac{x_i}{a_i} \right).$$

Substituting $p_i = a_i b_i$ into the right-hand side yields:

$$\sum_{i=1}^k p_i \frac{x_i}{a_i} = \sum_{i=1}^k (a_i b_i) \frac{x_i}{a_i} = \sum_{i=1}^k b_i x_i = b^T x.$$

Since $x \in VP(G)$, it follows that $b^T x \leq 1$. Consequently,

$$\log_2 \left(\sum_{i=1}^k p_i \frac{x_i}{a_i} \right) = \log_2(b^T x) \leq \log_2(1) = 0.$$

Expanding the left-hand side of the inequality yields:

$$\sum_{i=1}^k p_i \log_2 x_i - \sum_{i=1}^k p_i \log_2 a_i \leq 0,$$

which implies that for all $x \in VP(G)$:

$$-\sum_{i=1}^k p_i \log_2 a_i \leq -\sum_{i=1}^k p_i \log_2 x_i.$$

Therefore, the convex objective function $f(x) = -\sum_{i=1}^k p_i \log_2 x_i$ achieves its minimum over $VP(G)$ at $x = a$, and the value of $H(G, P)$ is attained by a . $\square$

Lemma 3.6. *Let $G = (V, E)$ be a graph with vertex set $|V| = n$. Then, we have:*

$$\left(\frac{1}{\chi_f(G)}, \frac{1}{\chi_f(G)}, \dots, \frac{1}{\chi_f(G)} \right)_{1 \times n} \in VP(G).$$

Proof. Let $\mathcal{I}(G) = \{S_1, S_2, \dots, S_m\}$ be the family of all independent sets of G , and $(w_1, w_2, \dots, w_m)$ be an optimal weight assignment attaining $\chi_f(G)$, so that $\sum_{j=1}^m w_j = \chi_f(G)$. For each independent set S_j , define the coefficient λ_j by:

$$\lambda_j = \frac{w_j}{\chi_f(G)}.$$

Since $w_j \geq 0$ and $\chi_f(G) > 0$, we have $\lambda_j \geq 0$ for all $j \in \{1, \dots, m\}$. Furthermore, the sum of these coefficients satisfies:

$$\sum_{j=1}^m \lambda_j = \sum_{j=1}^m \frac{w_j}{\chi_f(G)} = \frac{1}{\chi_f(G)} \sum_{j=1}^m w_j = \frac{\chi_f(G)}{\chi_f(G)} = 1.$$

Thus, the vector $\mathbf{x}$ defined as the convex combination of the characteristic vectors χ^{S_j} of the independent sets S_j :

$$\mathbf{x} = \sum_{j=1}^m \lambda_j \chi^{S_j}$$

is an element of $VP(G)$, i.e., $\mathbf{x} \in VP(G)$.

By definition of the characteristic vector χ^{S_j} , the i -th component of $\mathbf{x}$ evaluate as:

$$x_i = \sum_{j=1}^m \lambda_j (\chi^{S_j})_i = \sum_{j: v_i \in S_j} \lambda_j = \sum_{j: v_i \in S_j} \frac{w_j}{\chi_f(G)} = \frac{1}{\chi_f(G)} \sum_{j: v_i \in S_j} w_j.$$

By Definition 1.11, we obtain:

$$x_i \geq \frac{1}{\chi_f(G)} \quad \forall i \in \{1, \dots, n\}.$$

Since $VP(G)$ is a convex corner in $\mathbb{R}_+^n$, and following Definition 1.3, by setting $y_i = 1/\chi_f(G)$ for all $i \in \{1, \dots, n\}$, we have $0 \leq y_i \leq x_i$, which implies that:

$$\mathbf{y} = \left(\frac{1}{\chi_f(G)}, \frac{1}{\chi_f(G)}, \dots, \frac{1}{\chi_f(G)} \right) \in VP(G).$$

This completes the proof. $\square$

Lemma 3.7. *Let $G = (V, E)$ be a graph, and let $\{\lambda_{I_i}\}_{i=1}^t$ be an optimal weight assignment to its independent sets $\{I_i\}_{i=1}^t$ attaining the fractional chromatic number $\chi_f(G)$. For each vertex $v \in V(G)$, define its weight as:*

$$w(v) = \sum_{i=1}^t \lambda_{I_i}.$$

If there exists a vertex $v \in V(G)$ such that $w(v) > 1$, then removing v does not change the fractional chromatic number of G , i.e.,

$$\chi_f(G \setminus \{v\}) = \chi_f(G).$$

Proof. First, we show that $\chi_f(G \setminus \{v\}) \leq \chi_f(G)$. Suppose I is an independent set of G . It is clear that the set $I \setminus \{v\}$ is an independent set of both G and $G \setminus \{v\}$. Moreover, these sets constitute the complete family of independent sets of $G \setminus \{v\}$.

We assign the weights $\{\lambda'_{I_i \setminus \{v\}}\}_{i=1}^t$ to the independent sets of $G \setminus \{v\}$, such that:

$$\lambda'_{I \setminus \{v\}} = \begin{cases} \lambda_I + \lambda_{I \setminus \{v\}}, & \text{if } I \cup \{v\} \text{ independent in } G, \\ \lambda_{I \setminus \{v\}}, & \text{if } I \cup \{v\} \text{ not independent in } G. \end{cases}$$

Since the weight of each vertex here is equal to its previous weight, we get:

$$\chi_f(G \setminus \{v\}) \leq \sum_{J \in \mathcal{I}_G \setminus \{v\}} \lambda'_J = \sum_{i=1}^t \lambda_{I_i} = \chi_f(G).$$

Thus, removing a vertex does not increase the fractional chromatic number.

For the remainder of the proof, by contradiction, assume that:

$$\chi_f(G \setminus \{v\}) < \sum_{i=1}^t \lambda_{I_i}.$$

Hence, there must exist a weighting $\{\gamma_{I_i \setminus \{v\}}\}_{i=1}^t$ to the independent sets of $G \setminus \{v\}$, such that:

$$\sum_{J \in \mathcal{I}_G \setminus \{v\}} \gamma_J < \sum_{i=1}^t \lambda_{I_i}.$$

Now, using $\{\gamma_{I_i \setminus \{v\}}\}_{i=1}^t$, we define a new weighting of each independent set I_i of G , denoted by λ''_{I_i} , as follows:

$$\lambda''_{I_i} := \begin{cases} \frac{1}{w(v)} \lambda_{I_i} & (v \in I_i) \\ \frac{1}{w(v)} \lambda_{I_i} + \left(1 - \frac{1}{w(v)}\right) \gamma_{I_i \setminus \{v\}} & (v \notin I_i) \end{cases}$$

We will prove that the sum of the new weights is strictly less than $\sum_{i=1}^t \lambda_{I_i}$, for every i , $\lambda''_{I_i} \geq 0$, and that the new weight of every vertex is at least 1.

i. Verifying the non-negativity condition for independent set weights ($\lambda''_{I_i} \geq 0$):

- Case 1 ($v \in I_i$):

$$\lambda''_{I_i} = \frac{1}{w(v)} \lambda_{I_i} \geq 0$$

Since $w(v) > 1$ and $\lambda_{I_i} \geq 0$.

- Case 2 ($v \notin I_i$):

$$\lambda''_{I_i} = \frac{1}{w(v)} \lambda_{I_i} + \left(1 - \frac{1}{w(v)}\right) \gamma_{I_i \setminus \{v\}} \geq 0$$

Since $w(v) > 1$, the term $1 - \frac{1}{w(v)} > 0$, and all values of λ_i and $\gamma_{I_i \setminus \{v\}}$ are non-negative.

ii. Proving the vertex weight condition ($\sum_{u \in I_i} \lambda''_{I_i} \geq 1$):

- For vertex v :

$$\sum_{v \in I_i} \lambda''_{I_i} = \sum_{v \in I_i} \frac{1}{w(v)} \lambda_{I_i} = \frac{1}{w(v)} \sum_{v \in I_i} \lambda_{I_i} = \frac{1}{w(v)} \cdot w(v) = 1.$$

- For any vertex $u \neq v$:

$$\begin{aligned} \sum_{u \in I_i} \lambda''_{I_i} &= \sum_{\substack{u \in I_i \\ v \in I_i}} \frac{1}{w(v)} \lambda_{I_i} + \sum_{\substack{u \in I_i \\ v \notin I_i}} \left(\frac{1}{w(v)} \lambda_{I_i} + \left(1 - \frac{1}{w(v)}\right) \gamma_{I_i \setminus \{v\}} \right) \\ &= \frac{1}{w(v)} \sum_{u \in I_i} \lambda_{I_i} + \left(1 - \frac{1}{w(v)}\right) \sum_{\substack{u \in I_i \\ v \notin I_i}} \gamma_{I_i \setminus \{v\}} \\ &\geq \frac{1}{w(v)} (1) + \left(1 - \frac{1}{w(v)}\right) (1) = 1. \end{aligned}$$

iii. The sum of the new weights is strictly less than the initial weight sum ($\sum_{i=1}^t \lambda''_{I_i} < \sum_{i=1}^t \lambda_{I_i}$):

$$\begin{aligned} \sum_{i=1}^t \lambda''_{I_i} &= \sum_{v \in I_i} \frac{1}{w(v)} \lambda_{I_i} + \sum_{v \notin I_i} \left(\frac{1}{w(v)} \lambda_{I_i} + \left(1 - \frac{1}{w(v)}\right) \gamma_{I_i \setminus \{v\}} \right) \\ &= \sum_{i=1}^t \frac{1}{w(v)} \lambda_{I_i} + \sum_{v \notin I_i} \left(1 - \frac{1}{w(v)}\right) \gamma_{I_i \setminus \{v\}} \\ &< \sum_{i=1}^t \frac{1}{w(v)} \lambda_{I_i} + \left(1 - \frac{1}{w(v)}\right) \sum_{i=1}^t \lambda_{I_i} \quad (\text{since } \sum_{v \notin I_i} \gamma_{I_i \setminus \{v\}} < \sum_{i=1}^t \lambda_{I_i}) \\ &= \sum_{i=1}^t \lambda_{I_i}. \end{aligned}$$

As a result, the new weighting yields a smaller value.

Therefore, by the definition of the fractional chromatic number:

$$\chi_f(G) \neq \sum_{i=1}^t \lambda_{I_i},$$

which contradicts our initial assumption. □

Lemma 3.8. *G must have an induced subgraph G' such that $\chi_f(G') = \chi_f(G) =: \chi_f$, and for any $w \in VP(G')$ satisfying $\forall i: w_i \geq \frac{1}{\chi_f}$, we have equality for every i .*

Proof. In graph G , we iteratively remove every vertex whose weight is strictly greater than 1 under an optimal weighting for χ_f . By Lemma 3.7, removing these vertices does not change the fractional chromatic number. Eventually, we arrive at an induced subgraph G' of G such that for every optimal weighting of G' , the weight of every vertex is precisely equal to 1 (i.e., $w(x) = 1$).

We aim to prove that G' is indeed the desired induced subgraph in the lemma.

To this end, by contradiction, suppose there exists a vector $y = (y_1, y_2, \dots, y_m)$ in $VP(G')$ such that:

$$\forall 1 \leq i \leq m: \quad y_i \geq \frac{1}{\chi_f}$$

and at least one entry, say j , is strictly greater than $\frac{1}{\chi_f}$ ($y_j > \frac{1}{\chi_f}$).

Since y belongs to the packing polytope, it can be written as a convex combination of the characteristic vectors of independent sets of G' ; that is, there exist coefficients $\alpha_i \geq 0$ such that $\sum_{i=1}^{t'} \alpha_i = 1$. Now consider the coefficients $\{\alpha_i \chi_f\}_{i=1}^{t'}$ of independent sets. We will show that this forms an optimal weighting for the graph G' , yielding the fractional chromatic number χ_f .

The weight of every vertex in G' under this weighting is greater than or equal to 1, since for every $v_s \in V(G')$ we have:

$$w(v_s) = \sum_{v_s \in I_i} \alpha_i \chi_f = \chi_f \sum_{v_s \in I_i} \alpha_i = y_s \chi_f,$$

and by assumption $y_i \geq \frac{1}{\chi_f}$ for all entries, we have $y_s \geq \frac{1}{\chi_f}$, which implies:

$$w(v_s) = y_s \chi_f \geq 1.$$

On the other hand:

$$\sum_{i=1}^{t'} \alpha_i \chi_f = \chi_f \left(\sum_{i=1}^{t'} \alpha_i \right) = \chi_f.$$

Therefore, by the previous steps, $\{\alpha_i \chi_f\}_{i=1}^{t'}$ is an optimal weighting for G' . Thus, according to the initial assumption, the weight of every vertex must be equal to 1. However, for vertex v_j , we have:

$$w(v_j) = \sum_{v_j \in I_i} \alpha_i \chi_f = \chi_f \sum_{v_j \in I_i} \alpha_i = \chi_f y_j > \chi_f \left(\frac{1}{\chi_f} \right) = 1,$$

which is a contradiction. Consequently, no such vector y exists. $\square$

Theorem 3.9.

$$\max_Q H(G, Q) = \log_2 \chi_f(G),$$

where $\chi_f(G)$ is the fractional chromatic number of graph G .

Proof. Suppose that the maximum graph entropy of G is attained at the probability distribution $P = (p_1, p_2, \dots, p_n)$. By Lemma 3.6, and following from Definition 1.4, we have:

$$H(G, P) \leq \sum_{i=1}^n p_i \log_2 \left(\frac{1}{\frac{1}{\chi_f(G)}} \right) = \sum_{i=1}^n p_i \log_2 (\chi_f(G)) = \log_2 (\chi_f(G)) \sum_{i=1}^n p_i.$$

Since P is a valid probability distribution, we obtain:

$$\max_Q H(G, Q) = H(G, P) \leq \log_2 \chi_f(G).$$

It follows from Lemma 3.8 that G must contain an induced subgraph G' such that $\chi_f(G') = \chi_f(G)$, and for any $y \in VP(G')$ satisfying $y_i \geq \frac{1}{\chi_f(G)}$ for all i , equality holds for every i . Then, by Lemma 3.6 and 3.5, there exists a probability distribution P' on $V(G')$ such that:

$$H(G', P') = \sum_{i=1}^{|V(G')|} p'_i \log_2 \left(\frac{1}{\frac{1}{\chi_f(G)}} \right) = \log_2 \chi_f(G).$$

Extending this P' to a probability distribution P on $V(G)$ by setting $p_i = 0$ for all $i \in V(G) - V(G')$, we get a probability distribution attaining the right hand side in the statement:

$$\log_2 \chi_f(G) = H(G, P) \leq \max_Q H(G, Q).$$

Combining the two inequalities yields the desired equality, which completes the proof. $\square$

4 References

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